

# Western Mathematics

2024

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

## Mathematics Advanced

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### General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Approved calculators may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations

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**Total marks :  
100**

### **Section I – 10 marks** (pages 2 – 7)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

### **Section II – 90 marks** (pages 9 – 30)

- Attempt Questions 11 – 31
- Allow about 2 hours and 45 minutes for this section

## Section I

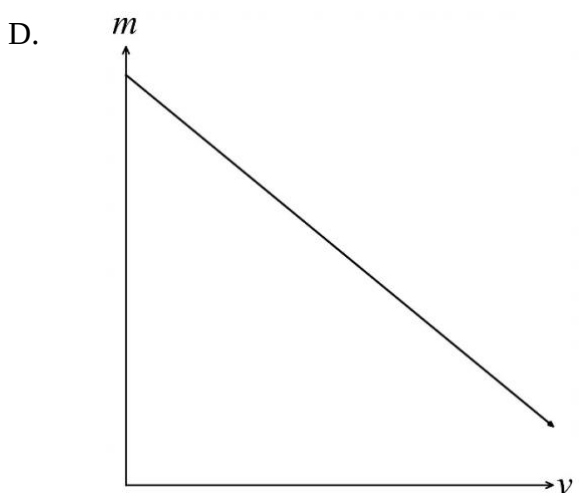
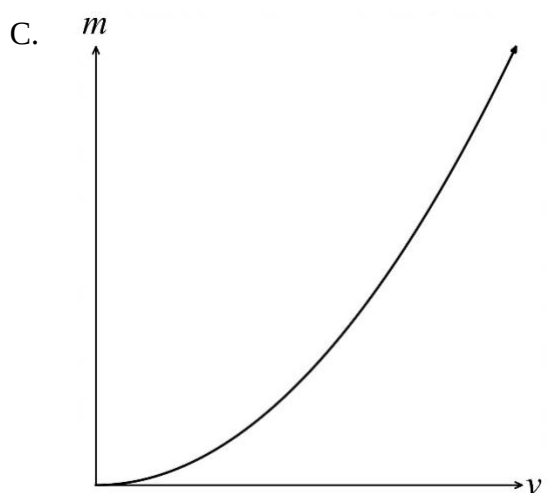
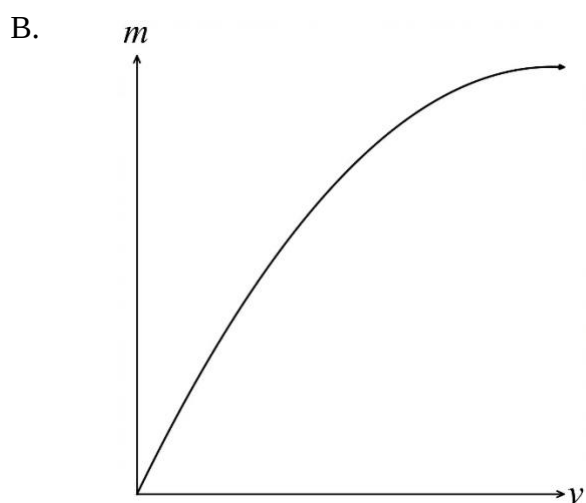
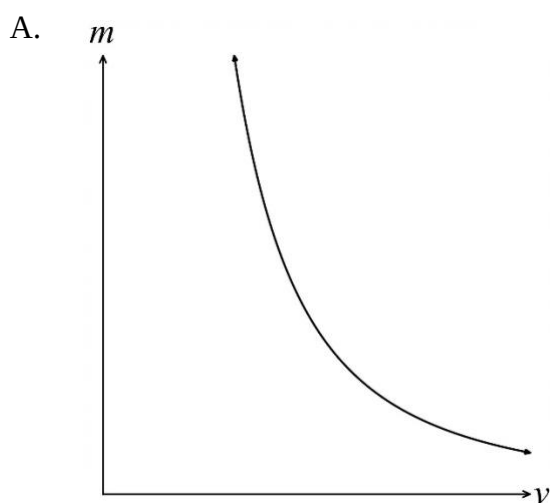
10 marks

Attempt Questions 1–10

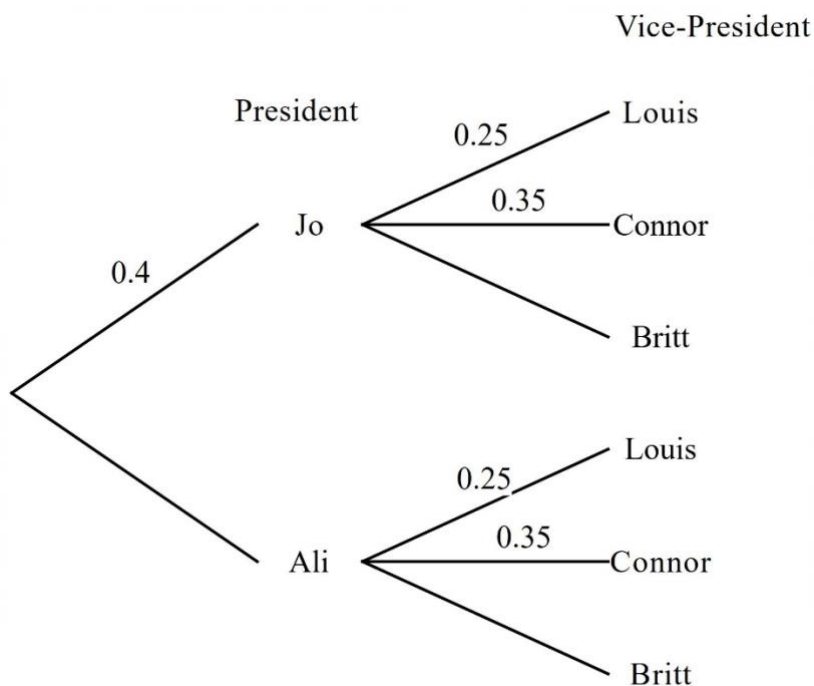
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10

1. It is known that a quantity  $m$  varies inversely with the square of a second quantity  $v$ . Which of these graphs could represent the relation between  $m$  and  $v$ ?



2. The partially completed probability tree below shows the voting intentions of shareholders in an election for President and Vice-President of a large corporation.



Based on this survey, what is the probability that Ali is elected President, with either Britt or Connor as Vice-President?

- A. 0.21
- B. 0.24
- C. 0.45
- D. 0.55
3. Consider the function  $f(x) = 4^{x+2}$ . What is the derivative of  $f(x)$  with respect to  $x$ ?
- A.  $2 \times 4^{x+2}$
- B.  $\ln 2 \times 4^{x+2}$
- C.  $\ln 4 \times 4^{x+2}$
- D.  $(x+2) \times 4^{x+2}$

4. The functions  $g(x)$  and  $h(x)$  are defined as follows:

$$g(x) = \frac{x^2}{8}$$

$$h(x) = \sqrt{2x} + 3$$

Which is the correct expression for  $h(g(x))$ ?

- A.  $\frac{x}{2} + 3$
- B.  $\frac{x}{2} - 3$
- C.  $\sqrt{\frac{2x}{3}}$
- D.  $\sqrt{\frac{x}{2}} + 3$
5. A circular Ferris wheel has a diameter of 40 metres and rotates at a constant rate, completing one full revolution every 2 minutes.

The height of a certain passenger above the ground as a function of time can be modelled by the equation:

$$h(t) = 20 \sin \left( \frac{\pi}{60}(t - 15) \right) + 30,$$

where  $h(t)$  represents the height in metres and  $t$  is the time in seconds.

What does the parameter 30 in the equation  $h(t) = 20 \sin \left( \frac{\pi}{60}(t - 15) \right) + 30$  represent in the context of the Ferris wheel?

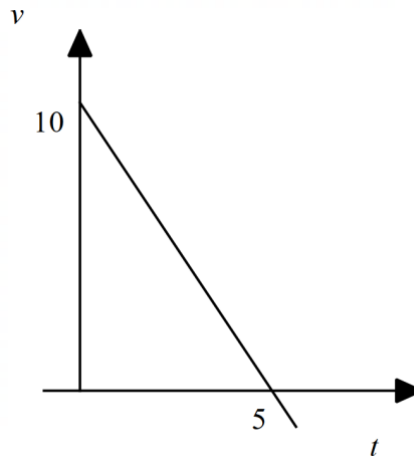
- A. The phase shift in time of the passenger on the Ferris wheel.
- B. The amplitude of the motion of the passenger on the Ferris wheel.
- C. The period of rotation of the Ferris wheel.
- D. The vertical shift of the centre of the Ferris wheel above the ground.

6. A particle is moving along the  $x$ -axis. The graph shows its velocity,  $v$  metres per second, at a time,  $t$  seconds.

When  $t = 0$ , the displacement  $x$  is equal to 3 metres.

What is the maximum value of the displacement  $x$ ?

- A. 10 m
- B. 13 m
- C. 28 m
- D. 53 m



7. A fitness tracker records the number of daily steps taken by users. Let  $X$  be the continuous random variable representing the daily step count, and  $f(x)$  be the probability density function. The probability density function is modelled as a uniform distribution over the interval  $[5000, 15000]$ , indicating that all step counts in this range are equally likely.

Using the properties of simple types of continuous random variables, what is the probability density function  $f(x)$  for this fitness tracker scenario?

- A.  $f(x) = \frac{1}{15\,000}$  for  $5\,000 \leq x \leq 15\,000$
- B.  $f(x) = \frac{1}{10\,000}$  for  $5\,000 \leq x \leq 15\,000$
- C.  $f(x) = \frac{1}{5\,000}$  for  $5\,000 \leq x \leq 15\,000$
- D.  $f(x) = \frac{1}{2}$  for  $5\,000 \leq x \leq 15\,000$

8. Find the derivative,  $g'(x)$ , of the function  $g(x) = \ln(\sqrt{x^2 - 1})$ .

A.  $g'(x) = 2x\sqrt{x^2 - 1}$

B.  $g'(x) = \frac{2x}{\sqrt{x^2 - 1}}$

C.  $g'(x) = \frac{2x}{x^2 - 1}$

D.  $g'(x) = \frac{x}{x^2 - 1}$

9. Simplify  $\frac{\sin^2 x + \sin x \cos x}{\cos^2 x + \sin x \cos x}$ .

A.  $\operatorname{cosec}^2 x$

B.  $\cot x$

C.  $\sec^2 x$

D.  $\tan x$

10. Data on the time in minutes it takes for students to complete a standardised test is normally distributed with a mean of 90 minutes and a standard deviation of 15 minutes.

Using the z-score table below, what is the approximate 75<sup>th</sup> percentile for the data?

| <b>z</b>   | <b>0.00</b> | <b>0.01</b> | <b>0.02</b> | <b>0.03</b> | <b>0.04</b> | <b>0.05</b> | <b>0.06</b> | <b>0.07</b> | <b>0.08</b> | <b>0.09</b> |
|------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| <b>0.0</b> | 0.5000      | 0.5040      | 0.5080      | 0.5120      | 0.5160      | 0.5199      | 0.5239      | 0.5279      | 0.5319      | 0.5359      |
| <b>0.1</b> | 0.5398      | 0.5438      | 0.5478      | 0.5517      | 0.5557      | 0.5596      | 0.5636      | 0.5675      | 0.5714      | 0.5753      |
| <b>0.2</b> | 0.5793      | 0.5832      | 0.5871      | 0.5910      | 0.5948      | 0.5987      | 0.6026      | 0.6064      | 0.6103      | 0.6141      |
| <b>0.3</b> | 0.6179      | 0.6217      | 0.6255      | 0.6293      | 0.6331      | 0.6368      | 0.6406      | 0.6443      | 0.6480      | 0.6517      |
| <b>0.4</b> | 0.6554      | 0.6591      | 0.6628      | 0.6664      | 0.6700      | 0.6736      | 0.6772      | 0.6808      | 0.6844      | 0.6879      |
| <b>0.5</b> | 0.6915      | 0.6950      | 0.6985      | 0.7019      | 0.7054      | 0.7088      | 0.7123      | 0.7157      | 0.7190      | 0.7224      |
| <b>0.6</b> | 0.7257      | 0.7291      | 0.7324      | 0.7357      | 0.7389      | 0.7422      | 0.7454      | 0.7486      | 0.7517      | 0.7549      |
| <b>0.7</b> | 0.7580      | 0.7611      | 0.7642      | 0.7673      | 0.7704      | 0.7734      | 0.7764      | 0.7794      | 0.7823      | 0.7852      |
| <b>0.8</b> | 0.7881      | 0.7910      | 0.7939      | 0.7967      | 0.7995      | 0.8023      | 0.8051      | 0.8078      | 0.8106      | 0.8133      |
| <b>0.9</b> | 0.8159      | 0.8186      | 0.8212      | 0.8238      | 0.8264      | 0.8289      | 0.8315      | 0.8340      | 0.8365      | 0.8389      |
| <b>1.0</b> | 0.8413      | 0.8438      | 0.8461      | 0.8485      | 0.8508      | 0.8531      | 0.8554      | 0.8577      | 0.8599      | 0.8621      |
| <b>1.1</b> | 0.8643      | 0.8665      | 0.8686      | 0.8708      | 0.8729      | 0.8749      | 0.8770      | 0.8790      | 0.8810      | 0.8830      |
| <b>1.2</b> | 0.8849      | 0.8869      | 0.8888      | 0.8907      | 0.8925      | 0.8944      | 0.8962      | 0.8980      | 0.8997      | 0.9015      |
| <b>1.3</b> | 0.9032      | 0.9049      | 0.9066      | 0.9082      | 0.9099      | 0.9115      | 0.9131      | 0.9147      | 0.9162      | 0.9177      |

- A. 95 minutes
- B. 100 minutes
- C. 105 minutes
- D. 110 minutes

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# Western Mathematics

2024 TRIAL HIGHER SCHOOL CERTIFICATE  
EXAMINATION

Class and Teacher

## Mathematics Advanced

Student Number

## Section II Answer Booklet

Student Name

90 marks

Attempt Questions 11-31

Allow about 2 hours and 45 minutes for this section

### Instructions

- Write your Name and Class details in the spaces provided at the top of this page.
- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of the booklet.  
If you use this space, clearly indicate which question you are answering.

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Please turn over

**Question 11** (2 marks)

In a role play game each player is randomly allocated a role by the game console at the start of each game.

**2**

The table below shows the probability of being allocated certain roles.

| Role Allocated      | Probability    |
|---------------------|----------------|
| Wizard              | $\frac{1}{20}$ |
| Knight              | $\frac{3}{20}$ |
| Sorceress           | $\frac{3}{40}$ |
| Amazon              | $\frac{9}{40}$ |
| Other passive Roles | $\frac{1}{2}$  |

Marielle likes to have the active roles of Knight or Amazon.

If she plays the game 64 times, how many times would she be expected to be allocated one of these two active roles?

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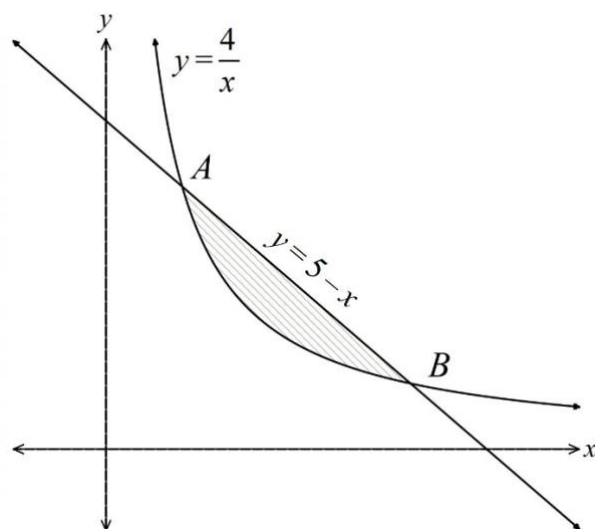
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**Question 12** (3 marks)

The curves  $y = 5 - x$  and  $y = \frac{4}{x}$  intersect at two points  $A$  and  $B$ , as shown below.

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Calculate the area between the two curves.

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**Question 13** (3 marks)

The first three terms of a series are 12, 18, 24.....

- (a) Calculate the tenth term in this series. **1**

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- (b) The sum of  $n$  consecutive terms of the series is 210. **2**

What is the value of  $n$ ?

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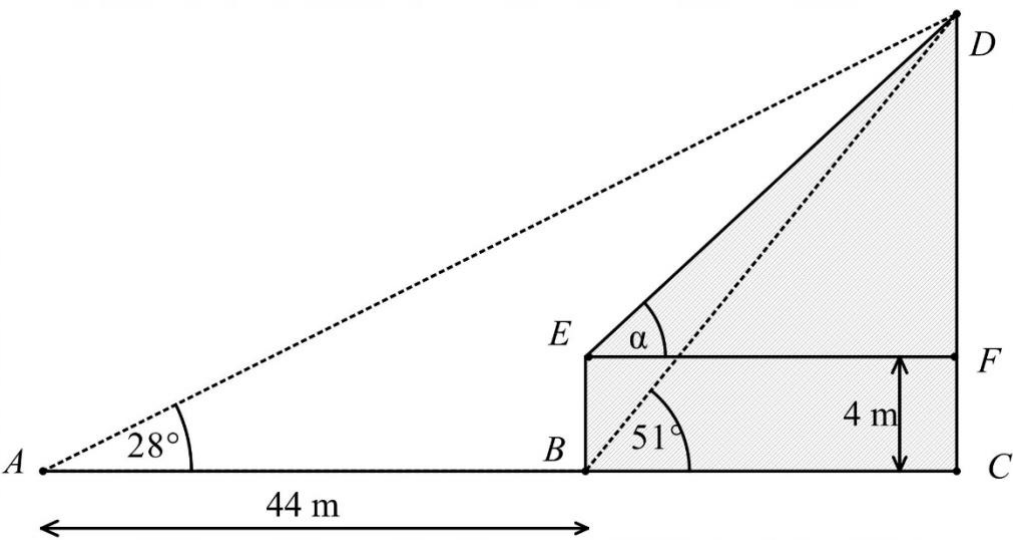
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**Question 14** (5 marks)

A grandstand at an oval is shown in cross-section  $BCDE$ .

The seating, along the interval  $DE$ , is angled at  $\alpha^\circ$  to the horizontal and begins 4 m above ground level.

From  $B$ , Aiden measures the angle of elevation of  $D$  to be  $51^\circ$ , then walks 44 m onto the field and measures the angle of elevation of  $D$  to be  $28^\circ$  from point  $A$ .



- (a) Calculate the distance  $BD$  correct to the nearest tenth of a metre. 2

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- (b) Calculate the distance  $DF$  and hence find the angle of the seating,  $\alpha$ , to the nearest degree. 3

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**Question 15** (4 marks)

Differentiate the following with respect to  $x$ :

(a)  $y = \ln (x^2 + 5x - 7)^2$  2

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(b)  $y = \frac{4\sin x}{\cos^2 x}$  2

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**Question 16** (2 marks)

Solve the equation  $2 \cos 2x = \sqrt{3}$  where  $0 \leq x \leq 2\pi$ .

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**Question 17** (4 marks)

Consider the geometric series

$$1 + (\sqrt{5} - 2) + (\sqrt{5} - 2)^2 + \dots$$

- (a) Show with calculations why this geometric series has a limiting sum.

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- (b) Find the exact value of the limiting sum, expressing your answer with a rational denominator.

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**Question 18** (3 marks)

Evaluate  $\int_0^{\frac{\pi}{2}} \sin\left(\frac{x}{2}\right) dx$

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**Question 19** (4 marks)

An object moves so that its distance ( $x$  metres) from the origin is given by:

$$x(t) = 6 - 2t + 8 \ln(t+3)$$

where time( $t$ ) is measured in seconds.

- (a) Find the position of the object when  $t = 0$ .

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- (b) Find expressions for  $x'(t)$  and  $x''(t)$  in terms of  $t$ .

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- (c) After how many seconds is  $x'(t) = 0$ ?

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**Question 20** (6 marks)

Natalie is managing the growth of a population of bacteria in a laboratory. The population can be modelled by the exponential equation  $P(t) = 100 e^{kt}$  where the initial population is 100 bacteria,  $k$  is the growth constant, and  $t$  is time in minutes.

- (a) Find the value of the growth constant  $k$  (correct to 1 significant figure), given that there are 3 660 bacteria present after 3 hours. 2

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- (b) Calculate the rate of change of the population with respect to time after 3 hours. (Answer to the nearest whole number) 2

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- (c) Natalie is concerned about the growth of the bacteria and wants to determine if she needs to introduce a control mechanism. 2

Use the derivatives,  $P'(t)$  and  $P''(t)$ , to show that the population is increasing at an increasing rate and hence determine if a control would be needed.

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**Question 21 (2 marks)**

Tables of future value and present value interest factors for an annuity of \$1 are shown below.

| Future Value Interest Factors for an Annuity of \$1 |                 |        |         |         |         |         |         |         |
|-----------------------------------------------------|-----------------|--------|---------|---------|---------|---------|---------|---------|
|                                                     | Rate per Period |        |         |         |         |         |         |         |
| Number of Periods                                   | 1.0%            | 2.0%   | 3.0%    | 4.0%    | 5.0%    | 6.0%    | 7.0%    | 8.0%    |
| 5                                                   | 5.101           | 5.204  | 5.309   | 5.416   | 5.526   | 5.637   | 5.751   | 5.867   |
| 10                                                  | 10.462          | 10.950 | 11.464  | 12.006  | 12.578  | 13.181  | 13.816  | 14.487  |
| 20                                                  | 22.019          | 24.297 | 26.870  | 29.778  | 33.066  | 36.786  | 40.995  | 45.762  |
| 30                                                  | 34.785          | 40.568 | 47.575  | 56.085  | 66.439  | 79.058  | 94.461  | 113.283 |
| 40                                                  | 48.886          | 60.402 | 75.401  | 95.026  | 120.800 | 154.762 | 199.635 | 259.057 |
| 50                                                  | 64.463          | 84.579 | 112.797 | 152.667 | 209.348 | 290.336 | 406.529 | 573.770 |

| Present Value Interest Factors for an Annuity of \$1 |                 |        |        |        |        |        |        |        |
|------------------------------------------------------|-----------------|--------|--------|--------|--------|--------|--------|--------|
|                                                      | Rate per Period |        |        |        |        |        |        |        |
| Number of Periods                                    | 1.0%            | 2.0%   | 3.0%   | 4.0%   | 5.0%   | 6.0%   | 7.0%   | 8.0%   |
| 5                                                    | 4.853           | 4.713  | 4.580  | 4.452  | 4.329  | 4.212  | 4.100  | 3.993  |
| 10                                                   | 9.471           | 8.983  | 8.530  | 8.111  | 7.722  | 7.360  | 7.024  | 6.710  |
| 20                                                   | 18.046          | 16.351 | 14.877 | 13.590 | 12.462 | 11.470 | 10.594 | 9.818  |
| 30                                                   | 25.808          | 22.396 | 19.600 | 17.292 | 15.372 | 13.765 | 12.409 | 11.258 |
| 40                                                   | 32.835          | 27.355 | 23.115 | 19.793 | 17.159 | 15.046 | 13.332 | 11.925 |
| 50                                                   | 39.196          | 31.424 | 25.730 | 21.482 | 18.256 | 15.762 | 13.801 | 12.233 |

- (a) Kumi plans to deposit \$5 000 at the end of each year into an annuity paying 8.0% p.a. compounding annually. What is the value of her investment after 10 years? **1**

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- (b) What amount could Kumi invest now at 8.0% p.a. compounding annually, to achieve the same financial outcome as the annuity after 10 years? **1**

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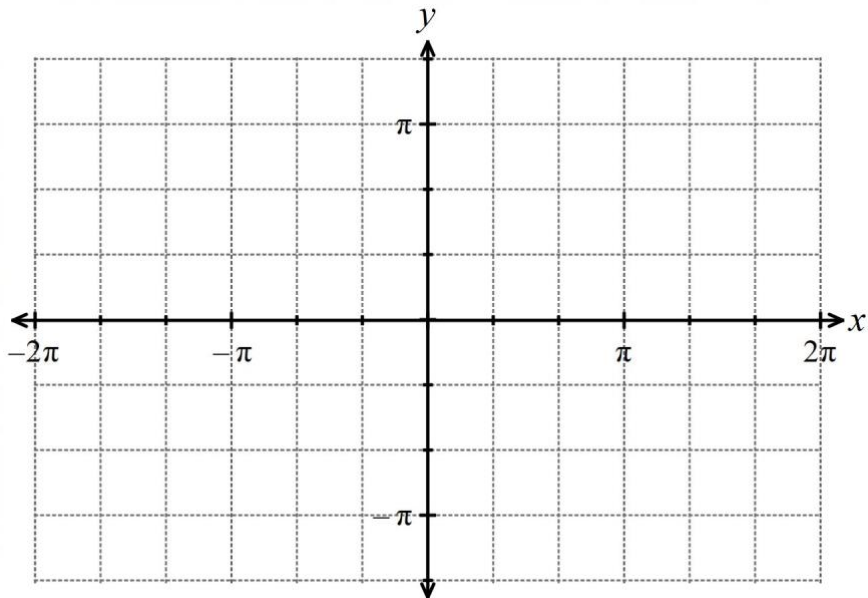
**Question 22** (4 marks)

- (a) On the grid below sketch the curves:

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$$y = \sqrt{\pi^2 - (x - \pi)^2}$$

$$y - x = 0$$



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- (b) Find the area enclosed between the two curves.

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**Question 23** (4 marks)

A company is analysing the productivity of its employees during a particular project.

The probability density function (PDF) represents the number of productive hours in the day of a random employee, and is modelled by the quadratic function:

$$f(x) = -\frac{3}{1\,372}(x^2 - 16x + 15)$$

where  $x$  is measured in hours.

- (a) Determine the modal number of the productive hours, denoted by  $M$ . 2

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- (b) Calculate the probability that an employee's productivity level is between 5 and 8 hours. 2

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**Question 24** (6 marks)

Sophia is designing a decorative garden fence, the top edge of which will follow a trigonometric curve.

The height of the top edge, in metres, is given by the equation  $h(x) = 0.5 \sin\left(2x - \frac{\pi}{2}\right) + 2$ ,  
where  $x$  represents the horizontal distance, in metres, along the fence.

The total length of the fence is  $2\pi$  metres.

- (a) Determine the amplitude, period, phase shift, and centre of the trigonometric function. 2

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- (b) Sophia wants to place supports at each point where the height of the fence is 2 metres. 2  
At which  $x$ -values in the range  $[0, 2\pi]$  does this occur?

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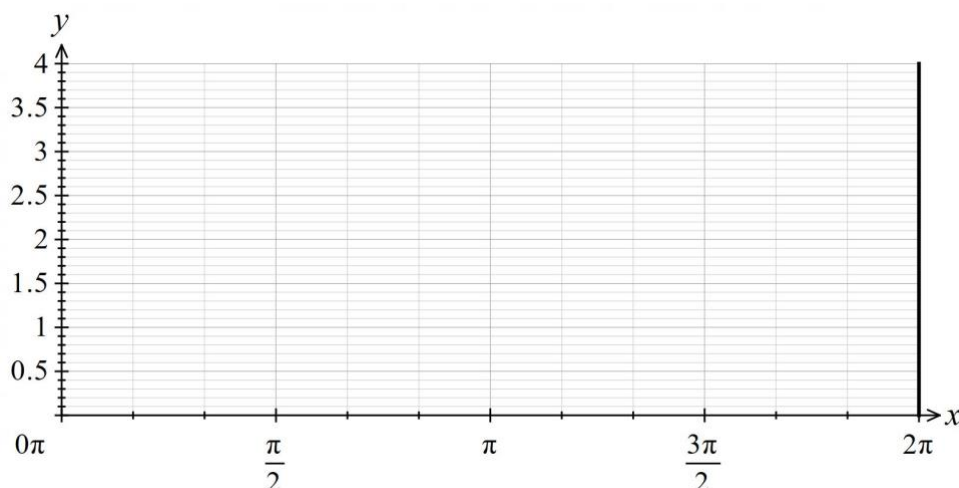
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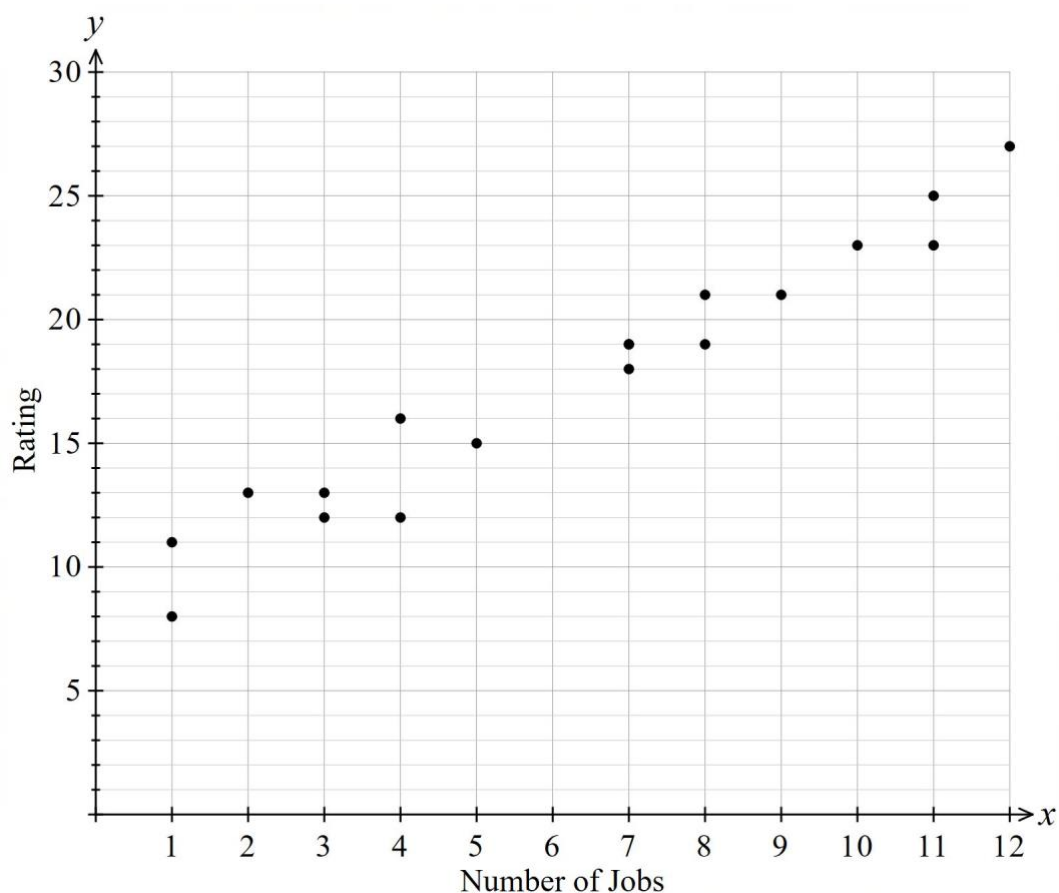
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- (c) Sketch the curve representing the top edge of the fence on the axes below. 2



**Question 25** (5 marks)

The scatterplot below shows the number of jobs that each apprentice in a mechanical engineering course has submitted, compared to the rating (out of 40) awarded to their most recent job.



Using  $x$  as the number of jobs and  $y$  as the most recent rating, the teacher finds the gradient of the least-squares regression line to be 1.5.

In any bivariate dataset, the least-squares regression line passes through the point  $(\bar{x}, \bar{y})$ , where  $\bar{x}$  is the sample mean of the  $x$ -values and  $\bar{y}$  is the sample mean of the  $y$ -values.

For this dataset,  $\bar{x} = 6$  and  $\bar{y} = 17$ .

- (a) Using the information above, determine the equation of the least-squares regression line and draw the line on the graph above. **3**

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**Question 25 continues on page 23**

Question 25 (continued)

- (b) Use the least-squares regression line to estimate the most recent rating for an apprentice who has submitted 16 jobs. 1

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- (c) Outline the limitations of using the least-squares regression line to estimate ratings for apprentices who have submitted much larger numbers of jobs. 1

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**Question 26** (2 marks)

Two tankers are releasing different chemicals into an empty holding tank in a manufacturing plant. 2

Tanker A is releasing its chemical at a rate of  $\left(2 + \frac{t^2}{t+1}\right)$  kilolitres per minute.

Tanker B is releasing its chemical at a rate of  $\left(1 + \frac{1}{t+1}\right)$  kilolitres per minute, where  $t$  is time in minutes for both expressions.

Show that the rate of flow of Tanker A is greater than the rate of flow of Tanker B by  $t$  kilolitres per minute.

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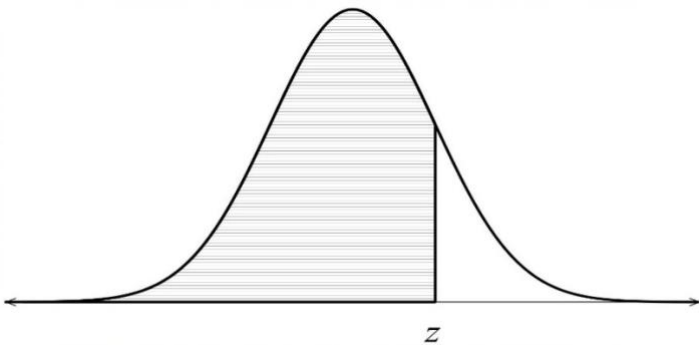
**Question 27** (5 marks)

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A random variable is normally distributed with a mean of 0 and a standard deviation of 1. The table gives the probability that this random variable lies below  $z$  for some positive values of  $z$ .

| $z$         | 0.5    | 0.6    | 0.7    | 0.8    | 0.9    | 1.0    | 1.1    | 1.2    | 1.3    | 1.4    | 1.5    |
|-------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Probability | 0.6915 | 0.7258 | 0.7580 | 0.7881 | 0.8159 | 0.8413 | 0.8643 | 0.8849 | 0.9032 | 0.9192 | 0.9332 |

The probability values given in the table are represented by the shaded area in the following diagram.



The marks on the entrance test for a scholarship are normally distributed with mean  $\mu = 125$  and standard deviation  $\sigma = 15$ .

All students who scored above 143 marks on the entrance test received a full scholarship and those who score between 137 and 143 received a partial scholarship.

There were 1 250 students who sat the examination. How many students received a partial scholarship?

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**Question 28** (7 marks)

Nick constructs a swimming pool in the shape below. Nick’s pool design is made up of a sector and a semi-circle, the radius( $r$ ) of the sector is equal to the diameter of the semi-circle.

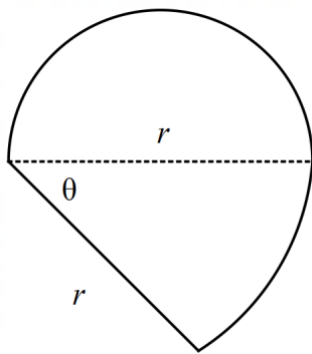


Figure not to scale

- (a) Find the perimeter of the pool in terms of  $r$  and  $\theta$ . 2

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- (b) Nick wants to be able to use this design in several different properties over the next year, so he decides to let the area of the pool equal 1 square unit for future ease in scaling the pool sizes. 2

Show that, if this is the case, the perimeter can be given by:

$$P = \frac{2}{r} + r \left( 1 + \frac{\pi}{4} \right).$$

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**Question 28 continues on page 26**

Question 28 (continued)

- (c) Show that the least perimeter occurs when  $r^2 = \frac{8}{\pi + 4}$  and calculate the corresponding value of  $\theta$  to the nearest degree.

3

[illegible]

**Question 29** (8 marks)

Let  $f(x) = e^{2x} (1 - x)$ .

A table of values is constructed below for  $[-3, 1]$

|        |      |      |      |      |      |
|--------|------|------|------|------|------|
| $x$    | -3   | -2   | -1   | 0    | 1    |
| $f(x)$ | 0.01 | 0.05 | 0.27 | 1.00 | 0.00 |

- (a) Find the coordinates of any stationary points, inflections and determine their nature for  $y = f(x)$ .4

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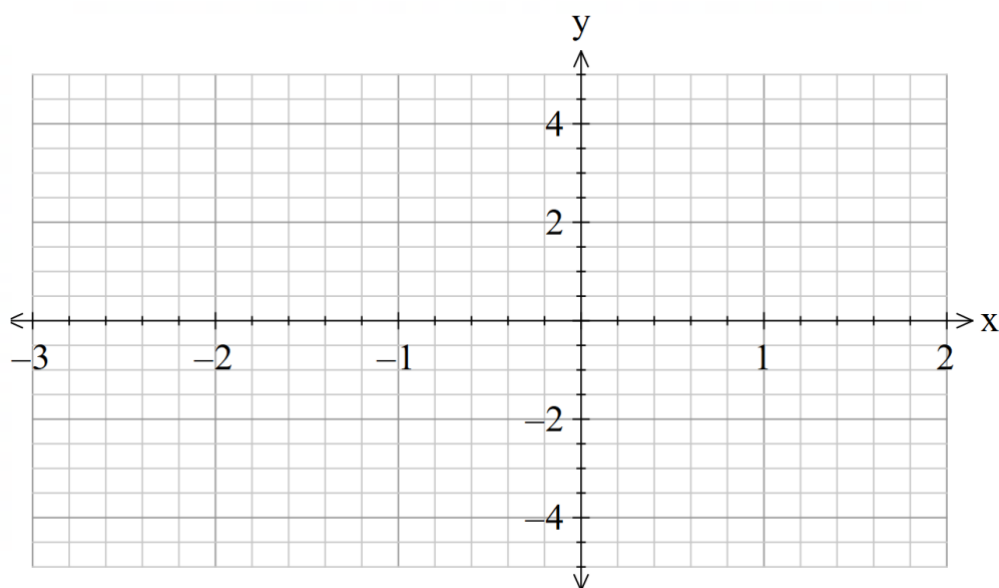
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**Question 29 continues on page 28**

Question 29 (continued)

- (b) Sketch the curve  $y = e^{2x} (1 - x)$  for  $[-3, 1]$ , showing the critical points.

2



- (c) Without any further calculus, state where, in the domain above, the curve is concave down.

1

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- (d) A student uses the trapezoidal rule with 5 function values to approximate the area under  $y = f(x)$  for  $[-3, 1]$  to be 1.325 units<sup>2</sup>.

1

By using your diagram above, state whether this approximation will be greater or less than the exact area under this curve, giving reasons for your answer.

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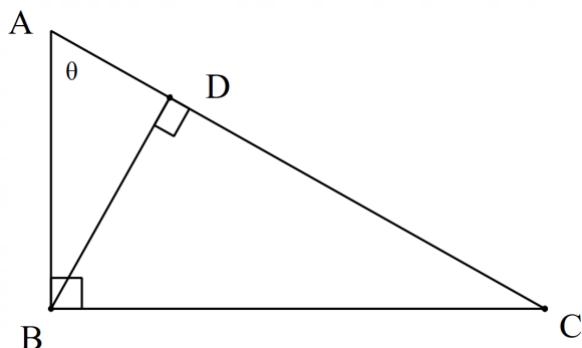
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**Question 30** (6 marks)

Emily constructed the diagram below using several right-angled triangles.

She wants to calculate the value  $\theta$ , measured in degrees, but needs help.

Jake tells her that he has determined that  $6AD + BC = 5AC$ .



- (a) Using the diagram and Jakes information, show how Emily arrives at the following equation:

2

$$6 \cos \theta + \tan \theta = 5 \sec \theta$$

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- (b) Hence or otherwise, show how the equation in part (a) could be rewritten in the following form:

2

$$6 \sin^2 \theta - \sin \theta - 1 = 0$$

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- (c) Find the value of  $\theta$ .

2

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**Question 31** (5 marks)

On Maisey’s 25<sup>th</sup> birthday she wins \$500 000 playing PowerScore. She invests her winnings on that day into a term deposit account paying 5% p.a. compounded monthly.

Every month, immediately after the interest is paid, Maisey withdraws \$4000 for her monthly expenses.

Before what birthday will Maisey’s winnings run out?

5

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[illegible][illegible]

Section II   Extra writing space

If you use this space, clearly indicate which question you are answering.

[illegible]



Section II   Extra writing space

If you use this space, clearly indicate which question you are answering.

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## Section II Extra writing space

If you use this space, clearly indicate which question you are answering.

[illegible]

Mathematics Advanced  
Mathematics Extension 1  
Mathematics Extension 2

REFERENCE SHEET

**Measurement****Length**

$$l = \frac{\theta}{360} \times 2\pi r$$

**Area**

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

**Surface area**

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

**Volume**

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

**Functions**

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For  $ax^3 + bx^2 + cx + d = 0$ :

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

**Relations**

$$(x - h)^2 + (y - k)^2 = r^2$$

**Financial Mathematics**

$$A = P(1 + r)^n$$

**Sequences and series**

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, r \neq 1$$

$$S = \frac{a}{1 - r}, |r| < 1$$

**Logarithmic and Exponential Functions**

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

## Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

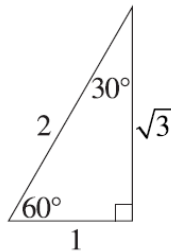
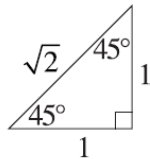
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$



### Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \quad \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \quad \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \quad \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

### Compound angles

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1+t^2}$$

$$\cos A = \frac{1-t^2}{1+t^2}$$

$$\tan A = \frac{2t}{1-t^2}$$

$$\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$$

$$\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$$

$$\sin A \cos B = \frac{1}{2}[\sin(A + B) + \sin(A - B)]$$

$$\cos A \sin B = \frac{1}{2}[\sin(A + B) - \sin(A - B)]$$

$$\sin^2 nx = \frac{1}{2}(1 - \cos 2nx)$$

$$\cos^2 nx = \frac{1}{2}(1 + \cos 2nx)$$

## Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

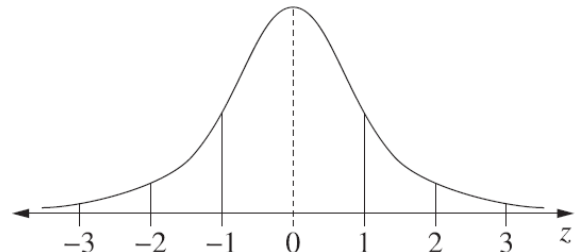
An outlier is a score

less than  $Q_1 - 1.5 \times IQR$

or

more than  $Q_3 + 1.5 \times IQR$

### Normal distribution



- approximately 68% of scores have z-scores between  $-1$  and  $1$
- approximately 95% of scores have z-scores between  $-2$  and  $2$
- approximately 99.7% of scores have z-scores between  $-3$  and  $3$

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

### Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

### Continuous random variables

$$P(X \leq r) = \int_a^r f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

### Binomial distribution

$$P(X = r) = {}^nC_r p^r (1-p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x)$$

$$= \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1-p)$$

## Differential Calculus

### Function

### Derivative

$$y = f(x)^n$$

$$\frac{dy}{dx} = n f'(x) [f(x)]^{n-1}$$

$$y = uv$$

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$y = g(u) \text{ where } u = f(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$y = \frac{u}{v}$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$y = \sin f(x)$$

$$\frac{dy}{dx} = f'(x) \cos f(x)$$

$$y = \cos f(x)$$

$$\frac{dy}{dx} = -f'(x) \sin f(x)$$

$$y = \tan f(x)$$

$$\frac{dy}{dx} = f'(x) \sec^2 f(x)$$

$$y = e^{f(x)}$$

$$\frac{dy}{dx} = f'(x) e^{f(x)}$$

$$y = \ln f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

$$y = a^{f(x)}$$

$$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$$

$$y = \log_a f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$$

$$y = \sin^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \cos^{-1} f(x)$$

$$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \tan^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{1 + [f(x)]^2}$$

## Integral Calculus

$$\int f'(x) [f(x)]^n dx = \frac{1}{n+1} [f(x)]^{n+1} + c$$

where  $n \neq -1$

$$\int f'(x) \sin f(x) dx = -\cos f(x) + c$$

$$\int f'(x) \cos f(x) dx = \sin f(x) + c$$

$$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

$$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_a^b f(x) dx$$

$$\approx \frac{b-a}{2n} \{f(a) + f(b) + 2[f(x_1) + \dots + f(x_{n-1})]\}$$

where  $a = x_0$  and  $b = x_n$

## Combinatorics

$${}^nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^nC_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \cdots + \binom{n}{r}x^{n-r}a^r + \cdots + a^n$$

---

## Vectors

$$|\underline{u}| = |x\underline{i} + y\underline{j}| = \sqrt{x^2 + y^2}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta = x_1x_2 + y_1y_2,$$

$$\text{where } \underline{u} = x_1\underline{i} + y_1\underline{j}$$

$$\text{and } \underline{v} = x_2\underline{i} + y_2\underline{j}$$

$$\underline{r} = \underline{a} + \lambda \underline{b}$$

---

## Complex Numbers

$$\begin{aligned} z = a + ib &= r(\cos \theta + i \sin \theta) \\ &= re^{i\theta} \end{aligned}$$

$$\begin{aligned} [r(\cos \theta + i \sin \theta)]^n &= r^n(\cos n\theta + i \sin n\theta) \\ &= r^n e^{in\theta} \end{aligned}$$

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## Mechanics

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left( \frac{1}{2} v^2 \right)$$

$$x = a \cos(nt + \alpha) + c$$

$$x = a \sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$

Western Mathematics  
**2024 Trial Higher School Certificate Examination**  
Mathematics Advanced

Name \_\_\_\_\_ Teacher \_\_\_\_\_

**Section I – Multiple Choice Answer Sheet**

**Allow about 15 minutes for this section.**

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

**Sample:**             $2 + 4 =$             (A) 2            (B) 6            (C) 8            (D) 9  
                                         A ☐            B ☒            C ☐            D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

                                         A ☒            B ☒            C ☐            D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

                                         A ☒            B ☒ <sup>correct</sup>            C ☐            D ☐

- |     |   |                       |   |                       |   |                       |   |                       |
|-----|---|-----------------------|---|-----------------------|---|-----------------------|---|-----------------------|
| 1.  | A | <input type="radio"/> | B | <input type="radio"/> | C | <input type="radio"/> | D | <input type="radio"/> |
| 2.  | A | <input type="radio"/> | B | <input type="radio"/> | C | <input type="radio"/> | D | <input type="radio"/> |
| 3.  | A | <input type="radio"/> | B | <input type="radio"/> | C | <input type="radio"/> | D | <input type="radio"/> |
| 4.  | A | <input type="radio"/> | B | <input type="radio"/> | C | <input type="radio"/> | D | <input type="radio"/> |
| 5.  | A | <input type="radio"/> | B | <input type="radio"/> | C | <input type="radio"/> | D | <input type="radio"/> |
| 6.  | A | <input type="radio"/> | B | <input type="radio"/> | C | <input type="radio"/> | D | <input type="radio"/> |
| 7.  | A | <input type="radio"/> | B | <input type="radio"/> | C | <input type="radio"/> | D | <input type="radio"/> |
| 8.  | A | <input type="radio"/> | B | <input type="radio"/> | C | <input type="radio"/> | D | <input type="radio"/> |
| 9.  | A | <input type="radio"/> | B | <input type="radio"/> | C | <input type="radio"/> | D | <input type="radio"/> |
| 10. | A | <input type="radio"/> | B | <input type="radio"/> | C | <input type="radio"/> | D | <input type="radio"/> |

Western Mathematics

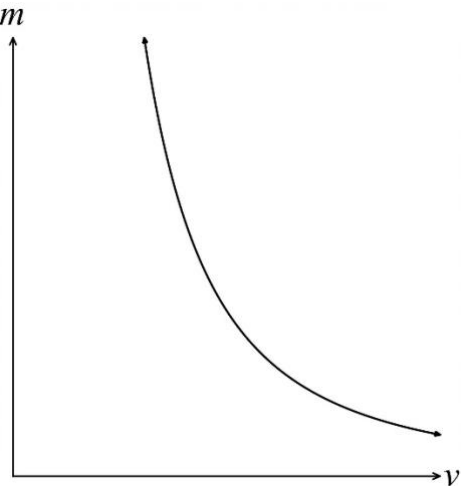
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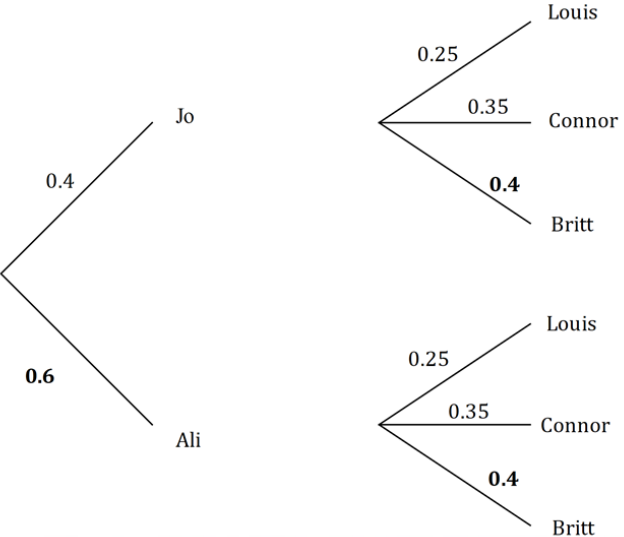
HSC TRIAL  
EXAMINATION

# Advanced Mathematics Solutions



## Section I

| No | Working                                                                                                                                                                                                                                                                                                                                                                             | Answer   |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1. | <p>As the quantities vary inversely, <math>m</math> will decrease as <math>v</math> increases in a hyperbolic shape, and as <math>m</math> varies inversely with the square of <math>v</math>, it will form a steeper curve than a simple hyperbola, so only option A meets these criteria.</p>  | <b>A</b> |

| No | Working                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Answer   |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2. | <div style="display: flex; justify-content: space-between;"> <div style="width: 45%;"> <p><math>P(\text{Ali is P}) = 1 - 0.4 = 0.6</math></p> <p><math>P(\text{Britt is VP}) = 1 - (0.25 + 0.35) = 0.4</math></p> <p><math>P(AC) = 0.6 \times 0.35 = 0.21</math></p> <p><math>P(AB) = 0.6 \times 0.4 = 0.24</math></p> <p><math>P(AC \text{ or } AB) = 0.21 + 0.24 = 0.45</math></p> </div> <div style="width: 45%; text-align: center;"> <p>President                      Vice-President</p>  </div> </div> | <b>C</b> |
| 3. | $f(x) = 4^{x+2}$ $f'(x) = \ln 4 \times 4^{x+2} \cdot 1$ $= \ln 4 \times 4^{x+2}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>C</b> |

| No | Working                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Answer   |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 4. | $g(x) = \frac{x^2}{8}$ $h(x) = \sqrt{2x} + 3$ $h(g(x)) = \sqrt{2(g(x))} + 3$ $= \sqrt{2\left(\frac{x^2}{8}\right)} + 3$ $= \sqrt{\frac{x^2}{4}} + 3$ $= \frac{x}{2} + 3$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>A</b> |
| 5. | <ul style="list-style-type: none"> <li>• The phase shift is the amount added to the variable <math>t</math>, so it is the passenger is 15 seconds out of phase</li> <li>• The amplitude is the coefficient of the sine function, so the amplitude is 20.</li> <li>• The period is given by <math>2\pi</math> divided by the coefficient of <math>t</math>, so <math>2\pi \div \frac{\pi}{60} = 120</math> (2 minutes as given)</li> <li>• The vertical shift is the constant added at the end, so the height of the centre of the wheel is 30.</li> </ul> <p>Therefore, the correct answer is:</p> <p>D) The vertical shift of the centre of the Ferris wheel.</p> | <b>D</b> |

| No | Working                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Answer   |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 6. | <div style="display: flex; justify-content: space-between;"> <div style="width: 45%;"> <p><b>Alternate solution 1</b></p> <p>Distance travelled = <math>\int_0^5 v dt</math></p> <p>= area under the velocity curve</p> <p><math>= \frac{1}{2} \times 5 \times 10</math></p> <p><math>= 25m</math></p> <p>Since velocity is always positive between <math>t = 0</math> and <math>t = 5</math> and the original displacement is <math>x = 3</math></p> <p>Max displacement</p> <p><math>= 25 + 3</math></p> <p><math>= 28m</math></p> </div> <div style="width: 45%;"> <p><b>Alternate solution 2</b></p> <p><math>\dot{x} = -2t + 10</math></p> <p><math>x = -t^2 + 10t + c</math></p> <p>when <math>t = 0, x = 3</math></p> <p><math>\therefore x = -t^2 + 10t + 3</math></p> <p>There is only one turning point, which is a maximum (concave down).</p> <p>This occurs when <math>v = 0</math>, which from the supplied graph is at <math>t = 5</math></p> <p>when <math>t = 5</math></p> <p><math>x = -5^2 + 10(5) + 3</math></p> <p><math>= -25 + 50 + 3</math></p> <p><math>= 28</math></p> </div> </div> | <b>C</b> |
| 7. | <p>For a uniform distribution, the probability density function <math>f(x)</math> is constant within the interval <math>[a, b]</math> and is given by</p> $f(x) = \frac{1}{b-a}$ <p>In this case, the interval is <math>[5000, 15000]</math>, so the correct answer is:</p> <p>A) <math>f(x) = \frac{1}{10\,000}</math> for <math>5\,000 \leq x \leq 15\,000</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>B</b> |

| No | Working                                                                                                                                                                                                                                                                                                                                                                                                                                                | Answer   |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 8. | $g(x) = \ln(\sqrt{x^2 - 1}) = \ln\left((x^2 - 1)^{\frac{1}{2}}\right)$ $g'(x) = \frac{\frac{d}{dx}\left((x^2 - 1)^{\frac{1}{2}}\right)}{(x^2 - 1)^{\frac{1}{2}}}$ $= \frac{\frac{1}{2}\left((x^2 - 1)^{-\frac{1}{2}}\right) \times 2x}{(x^2 - 1)^{\frac{1}{2}}}$ $= \frac{x\left((x^2 - 1)^{-\frac{1}{2}}\right)}{(x^2 - 1)^{\frac{1}{2}}}$ $= \frac{x}{(x^2 - 1)^{\frac{1}{2}}} \times \frac{1}{(x^2 - 1)^{\frac{1}{2}}}$ $g'(x) = \frac{x}{x^2 - 1}$ | <b>D</b> |
| 9. | $\frac{\sin^2 x + \sin x \cos x}{\cos^2 x + \sin x \cos x}$ $= \frac{\sin x(\sin x + \cos x)}{\cos x(\sin x + \cos x)}$ $= \frac{\sin x}{\cos x}$ $= \tan x$                                                                                                                                                                                                                                                                                           | <b>D</b> |

| No  | Working                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Answer |        |        |        |        |        |        |        |        |      |      |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |          |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|------|------|-----|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|-----|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|-----|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|-----|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|-----|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|-----|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|-----|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|-----|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|-----|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|-----|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|-----|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|-----|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|-----|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|-----|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|----------|
| 10. | <p>The 75th percentile corresponds to the point where 75% of the data falls below and 25% falls above. Using the normal distribution properties, we can find this value by locating it on table of z-scores and then converting it back to the original scale.</p> <p>For a standard normal distribution, the z-score corresponding to the 75th percentile is approximately 0.675.</p> <table><tr><th>z</th><th>0.00</th><th>0.01</th><th>0.02</th><th>0.03</th><th>0.04</th><th>0.05</th><th>0.06</th><th>0.07</th><th>0.08</th><th>0.09</th></tr><tr><td>0.0</td><td>0.5000</td><td>0.5040</td><td>0.5080</td><td>0.5120</td><td>0.5160</td><td>0.5199</td><td>0.5239</td><td>0.5279</td><td>0.5319</td><td>0.5359</td></tr><tr><td>0.1</td><td>0.5398</td><td>0.5438</td><td>0.5478</td><td>0.5517</td><td>0.5557</td><td>0.5596</td><td>0.5636</td><td>0.5675</td><td>0.5714</td><td>0.5753</td></tr><tr><td>0.2</td><td>0.5793</td><td>0.5832</td><td>0.5871</td><td>0.5910</td><td>0.5948</td><td>0.5987</td><td>0.6026</td><td>0.6064</td><td>0.6103</td><td>0.6141</td></tr><tr><td>0.3</td><td>0.6179</td><td>0.6217</td><td>0.6255</td><td>0.6293</td><td>0.6331</td><td>0.6368</td><td>0.6406</td><td>0.6443</td><td>0.6480</td><td>0.6517</td></tr><tr><td>0.4</td><td>0.6554</td><td>0.6591</td><td>0.6628</td><td>0.6664</td><td>0.6700</td><td>0.6736</td><td>0.6772</td><td>0.6808</td><td>0.6844</td><td>0.6879</td></tr><tr><td>0.5</td><td>0.6915</td><td>0.6950</td><td>0.6985</td><td>0.7019</td><td>0.7054</td><td>0.7088</td><td>0.7123</td><td>0.7157</td><td>0.7190</td><td>0.7224</td></tr><tr><td>0.6</td><td>0.7257</td><td>0.7291</td><td>0.7324</td><td>0.7357</td><td>0.7389</td><td>0.7422</td><td>0.7454</td><td>0.7486</td><td>0.7517</td><td>0.7549</td></tr><tr><td>0.7</td><td>0.7580</td><td>0.7611</td><td>0.7642</td><td>0.7673</td><td>0.7704</td><td>0.7734</td><td>0.7764</td><td>0.7794</td><td>0.7823</td><td>0.7852</td></tr><tr><td>0.8</td><td>0.7881</td><td>0.7910</td><td>0.7939</td><td>0.7967</td><td>0.7995</td><td>0.8023</td><td>0.8051</td><td>0.8078</td><td>0.8106</td><td>0.8133</td></tr><tr><td>0.9</td><td>0.8159</td><td>0.8186</td><td>0.8212</td><td>0.8238</td><td>0.8264</td><td>0.8289</td><td>0.8315</td><td>0.8340</td><td>0.8365</td><td>0.8389</td></tr><tr><td>1.0</td><td>0.8413</td><td>0.8438</td><td>0.8461</td><td>0.8485</td><td>0.8508</td><td>0.8531</td><td>0.8554</td><td>0.8577</td><td>0.8599</td><td>0.8621</td></tr><tr><td>1.1</td><td>0.8643</td><td>0.8665</td><td>0.8686</td><td>0.8708</td><td>0.8729</td><td>0.8749</td><td>0.8770</td><td>0.8790</td><td>0.8810</td><td>0.8830</td></tr><tr><td>1.2</td><td>0.8849</td><td>0.8869</td><td>0.8888</td><td>0.8907</td><td>0.8925</td><td>0.8944</td><td>0.8962</td><td>0.8980</td><td>0.8997</td><td>0.9015</td></tr><tr><td>1.3</td><td>0.9032</td><td>0.9049</td><td>0.9066</td><td>0.9082</td><td>0.9099</td><td>0.9115</td><td>0.9131</td><td>0.9147</td><td>0.9162</td><td>0.9177</td></tr></table> <p>Using the formula <math>z = \frac{X - \mu}{\sigma}</math>, where <math>\mu</math> is the mean, <math>z</math> is the z-score, and <math>\sigma</math> is the standard deviation, we get:</p> $X = 90 + 0.675 \times 15$ $\approx 100.125$ <p><math>\therefore</math> closest option is 100</p> | z      | 0.00   | 0.01   | 0.02   | 0.03   | 0.04   | 0.05   | 0.06   | 0.07   | 0.08 | 0.09 | 0.0 | 0.5000 | 0.5040 | 0.5080 | 0.5120 | 0.5160 | 0.5199 | 0.5239 | 0.5279 | 0.5319 | 0.5359 | 0.1 | 0.5398 | 0.5438 | 0.5478 | 0.5517 | 0.5557 | 0.5596 | 0.5636 | 0.5675 | 0.5714 | 0.5753 | 0.2 | 0.5793 | 0.5832 | 0.5871 | 0.5910 | 0.5948 | 0.5987 | 0.6026 | 0.6064 | 0.6103 | 0.6141 | 0.3 | 0.6179 | 0.6217 | 0.6255 | 0.6293 | 0.6331 | 0.6368 | 0.6406 | 0.6443 | 0.6480 | 0.6517 | 0.4 | 0.6554 | 0.6591 | 0.6628 | 0.6664 | 0.6700 | 0.6736 | 0.6772 | 0.6808 | 0.6844 | 0.6879 | 0.5 | 0.6915 | 0.6950 | 0.6985 | 0.7019 | 0.7054 | 0.7088 | 0.7123 | 0.7157 | 0.7190 | 0.7224 | 0.6 | 0.7257 | 0.7291 | 0.7324 | 0.7357 | 0.7389 | 0.7422 | 0.7454 | 0.7486 | 0.7517 | 0.7549 | 0.7 | 0.7580 | 0.7611 | 0.7642 | 0.7673 | 0.7704 | 0.7734 | 0.7764 | 0.7794 | 0.7823 | 0.7852 | 0.8 | 0.7881 | 0.7910 | 0.7939 | 0.7967 | 0.7995 | 0.8023 | 0.8051 | 0.8078 | 0.8106 | 0.8133 | 0.9 | 0.8159 | 0.8186 | 0.8212 | 0.8238 | 0.8264 | 0.8289 | 0.8315 | 0.8340 | 0.8365 | 0.8389 | 1.0 | 0.8413 | 0.8438 | 0.8461 | 0.8485 | 0.8508 | 0.8531 | 0.8554 | 0.8577 | 0.8599 | 0.8621 | 1.1 | 0.8643 | 0.8665 | 0.8686 | 0.8708 | 0.8729 | 0.8749 | 0.8770 | 0.8790 | 0.8810 | 0.8830 | 1.2 | 0.8849 | 0.8869 | 0.8888 | 0.8907 | 0.8925 | 0.8944 | 0.8962 | 0.8980 | 0.8997 | 0.9015 | 1.3 | 0.9032 | 0.9049 | 0.9066 | 0.9082 | 0.9099 | 0.9115 | 0.9131 | 0.9147 | 0.9162 | 0.9177 | <b>B</b> |
| z   | 0.00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0.01   | 0.02   | 0.03   | 0.04   | 0.05   | 0.06   | 0.07   | 0.08   | 0.09   |      |      |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |          |
| 0.0 | 0.5000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0.5040 | 0.5080 | 0.5120 | 0.5160 | 0.5199 | 0.5239 | 0.5279 | 0.5319 | 0.5359 |      |      |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |          |
| 0.1 | 0.5398                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0.5438 | 0.5478 | 0.5517 | 0.5557 | 0.5596 | 0.5636 | 0.5675 | 0.5714 | 0.5753 |      |      |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |          |
| 0.2 | 0.5793                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0.5832 | 0.5871 | 0.5910 | 0.5948 | 0.5987 | 0.6026 | 0.6064 | 0.6103 | 0.6141 |      |      |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |          |
| 0.3 | 0.6179                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0.6217 | 0.6255 | 0.6293 | 0.6331 | 0.6368 | 0.6406 | 0.6443 | 0.6480 | 0.6517 |      |      |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |          |
| 0.4 | 0.6554                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0.6591 | 0.6628 | 0.6664 | 0.6700 | 0.6736 | 0.6772 | 0.6808 | 0.6844 | 0.6879 |      |      |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |          |
| 0.5 | 0.6915                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0.6950 | 0.6985 | 0.7019 | 0.7054 | 0.7088 | 0.7123 | 0.7157 | 0.7190 | 0.7224 |      |      |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |          |
| 0.6 | 0.7257                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0.7291 | 0.7324 | 0.7357 | 0.7389 | 0.7422 | 0.7454 | 0.7486 | 0.7517 | 0.7549 |      |      |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |          |
| 0.7 | 0.7580                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0.7611 | 0.7642 | 0.7673 | 0.7704 | 0.7734 | 0.7764 | 0.7794 | 0.7823 | 0.7852 |      |      |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |          |
| 0.8 | 0.7881                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0.7910 | 0.7939 | 0.7967 | 0.7995 | 0.8023 | 0.8051 | 0.8078 | 0.8106 | 0.8133 |      |      |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |          |
| 0.9 | 0.8159                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0.8186 | 0.8212 | 0.8238 | 0.8264 | 0.8289 | 0.8315 | 0.8340 | 0.8365 | 0.8389 |      |      |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |          |
| 1.0 | 0.8413                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0.8438 | 0.8461 | 0.8485 | 0.8508 | 0.8531 | 0.8554 | 0.8577 | 0.8599 | 0.8621 |      |      |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |          |
| 1.1 | 0.8643                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0.8665 | 0.8686 | 0.8708 | 0.8729 | 0.8749 | 0.8770 | 0.8790 | 0.8810 | 0.8830 |      |      |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |          |
| 1.2 | 0.8849                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0.8869 | 0.8888 | 0.8907 | 0.8925 | 0.8944 | 0.8962 | 0.8980 | 0.8997 | 0.9015 |      |      |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |          |
| 1.3 | 0.9032                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0.9049 | 0.9066 | 0.9082 | 0.9099 | 0.9115 | 0.9131 | 0.9147 | 0.9162 | 0.9177 |      |      |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |     |        |        |        |        |        |        |        |        |        |        |          |

Western Mathematics  
**2024 Trial Higher School Certificate Examination**  
Mathematics Advanced

Name \_\_\_\_\_ Teacher \_\_\_\_\_

**Section I – Multiple Choice Answer Sheet**

**Allow about 15 minutes for this section**

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

**Sample:**             $2 + 4 =$             (A) 2            (B) 6            (C) 8            (D) 9  
                                         A ☐            B ☒            C ☐            D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

                                         A ☒            B ☒            C ☐            D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

                                         A ☒            B ☒ <sup>correct</sup>            C ☐            D ☐

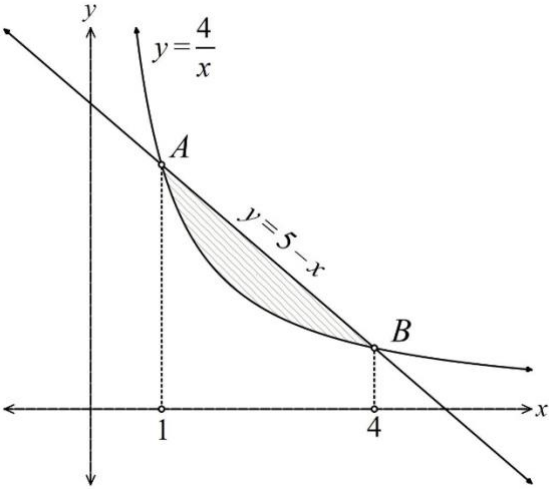
- |     |                                    |                                    |                                    |                                    |
|-----|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| 1.  | A <input checked="" type="radio"/> | B <input type="radio"/>            | C <input type="radio"/>            | D <input type="radio"/>            |
| 2.  | A <input type="radio"/>            | B <input type="radio"/>            | C <input checked="" type="radio"/> | D <input type="radio"/>            |
| 3.  | A <input type="radio"/>            | B <input type="radio"/>            | C <input checked="" type="radio"/> | D <input type="radio"/>            |
| 4.  | A <input checked="" type="radio"/> | B <input type="radio"/>            | C <input type="radio"/>            | D <input type="radio"/>            |
| 5.  | A <input type="radio"/>            | B <input type="radio"/>            | C <input type="radio"/>            | D <input checked="" type="radio"/> |
| 6.  | A <input type="radio"/>            | B <input type="radio"/>            | C <input checked="" type="radio"/> | D <input type="radio"/>            |
| 7.  | A <input type="radio"/>            | B <input checked="" type="radio"/> | C <input type="radio"/>            | D <input type="radio"/>            |
| 8.  | A <input type="radio"/>            | B <input type="radio"/>            | C <input type="radio"/>            | D <input checked="" type="radio"/> |
| 9.  | A <input type="radio"/>            | B <input type="radio"/>            | C <input type="radio"/>            | D <input checked="" type="radio"/> |
| 10. | A <input type="radio"/>            | B <input checked="" type="radio"/> | C <input type="radio"/>            | D <input type="radio"/>            |

Solutions 2024

Section II

| Question | Working and answer                                                                                                                                                                           | Marks | Mark Allocation                                                                                          |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------------------------------------------------------------------------|
| 11.      | $\begin{aligned} P(\text{active role}) &= \frac{3}{20} + \frac{9}{40} \\ &= \frac{3}{8} \\ \text{Expected frequency} &= np \\ &= 64 \times \frac{3}{8} \\ &= 24 \text{ times} \end{aligned}$ | 2     | <p>2 marks for correct solution</p><br><br><br><p>1 mark for correct probability or equivalent merit</p> |



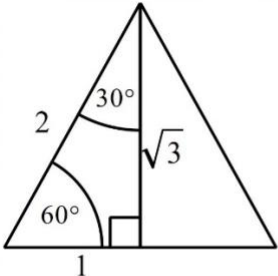
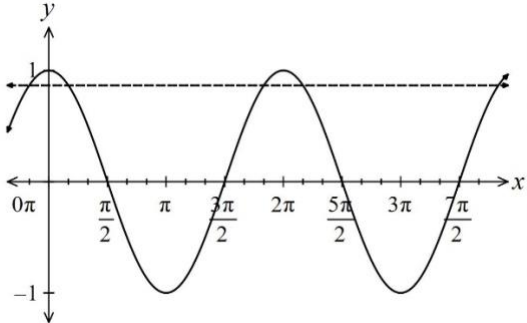
| Question | Working and answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks    | Mark Allocation                                                                                                                                                                                                                                                                     |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12.      | <p>Intersection</p> $5 - x = \frac{4}{x}$ $x(5 - x) = 4$ $5x - x^2 = 4$ $x^2 - 5x + 4 = 0$ $(x - 4)(x - 1) = 0$ $x = 4 \text{ or } 1$ <p>when <math>x = 4</math> <math>y = 1</math></p> <p><math>x = 1</math> <math>y = 4</math></p>  $\int_1^4 5 - x \, dx - \int_1^4 \frac{4}{x} \, dx$ $= \left[ 5x - \frac{x^2}{2} \right]_1^4 - [4 \ln x]_1^4$ $= \left[ (20 - 8) - \left( 5 - \frac{1}{2} \right) \right] - (4 \ln 4 - 4 \ln 1)$ $= \frac{15}{2} - 4 \ln 4 \text{ units}^2$ | <b>3</b> | <p>3 marks for correct solution</p> <p>2 marks for finding the intersection points and progress towards finding the area between the curves or equivalent merit</p> <p>1 mark for finding the intersection or trying to find an area between the two curves or equivalent merit</p> |





| Question |    | Working and answer                                                                                                       | Marks    | Mark Allocation                                                                                                                        |
|----------|----|--------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------|
| 15.      | a) | $y = \ln (x^2 + 5x - 7)^2$ $y' = \frac{2(2x + 5)(x^2 + 5x - 7)}{(x^2 + 5x - 7)^2}$ $y' = \frac{2(2x + 5)}{x^2 + 5x - 7}$ | <b>2</b> | <p>2 marks for correct solution</p> <p>1 mark for differentiation using <math>\frac{f'(x)}{f(x)}</math></p> <p>or equivalent merit</p> |

| Question |    | Working and answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks | Mark Allocation                                                                                                   |
|----------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------------------------------------|
| 15.      | b) | $y = \frac{4\sin x}{\cos^2 x}$ $u = 4\sin x \quad v = \cos^2 x$ $u' = 4\cos x \quad v' = -2\sin x \cos x$ $y' = \frac{4\cos^3 x + 8\sin^2 x \cos x}{\cos^4 x}$ <p>Full marks for achieving solution to this point</p> <p>Any incorrect working past this point can be ignored</p> <p>Full marks for any of the following or for further simplification beyond these</p> $y' = \frac{4\cos^2 x + 8\sin^2 x}{\cos^3 x}$ $y' = \frac{4\cos x(\cos^2 x + 2\sin^2 x)}{\cos^4 x}$ $y' = \frac{4(\cos^2 x + 2\sin^2 x)}{\cos^3 x}$ $y' = \frac{4(1 + \sin^2 x)}{\cos^3 x}$ | 2     | <p>2 marks for correct solution</p> <p>1 mark for differentiation using the quotient rule or equivalent merit</p> |

| Question | Working and answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks | Mark Allocation                                                                                                                       |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------|
| 16.      | $2\cos 2x = \sqrt{3}$ $\cos 2x = \frac{\sqrt{3}}{2}$ <p>cos +ve 1st, 4th quadrants</p>   $\cos 30^\circ = \cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$ $2x = \frac{\pi}{6}, \frac{11\pi}{6}, \frac{13\pi}{6}, \frac{23\pi}{6} \text{ for } [0, 4\pi]$ $x = \frac{\pi}{12}, \frac{11\pi}{12}, \frac{13\pi}{12}, \frac{23\pi}{12} \text{ for } [0, 2\pi]$ | 2     | <p>2 marks for all correct solutions</p> <p>1 mark for some correct values or at least finding 2 values of x, or equivalent merit</p> |

| Question |    | Working and answer                                                                                                                                                                                                                                  | Marks | Mark Allocation                                                                                                                                                                                                                                                                                                                                                                 |
|----------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 17.      | a) | <p>For a geometric series to have a limiting sum <math> r  &lt; 1</math></p> $\left  \frac{T_2}{T_1} \right  < 1$ $\frac{\sqrt{5} - 2}{1} \approx 0.24$ <p><math>\therefore  r  &lt; 1</math></p> <p><math>\therefore</math> has a limiting sum</p> | 2     | <p>2 marks for correct explanation including that a limiting sum occurs if <math> r  &lt; 1</math> and showing that for this series <math> r  &lt; 1</math></p> <p>1 mark for stating that a limiting sum occurs if <math> r  &lt; 1</math> then error in algebra, or for simply showing that for this series <math>r &lt; 1</math> without explanation or equivalent merit</p> |

| Question |    | Working and answer                                                                                                                                                                 | Marks | Mark Allocation                                                                                                                 |
|----------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------|
| 17.      | b) | $S = \frac{a}{1-r}$ $= \frac{1}{1 - (\sqrt{5} - 2)}$ $= \frac{1}{3 - \sqrt{5}} \times \frac{3 + \sqrt{5}}{3 + \sqrt{5}}$ $= \frac{3 + \sqrt{5}}{9 - 5}$ $= \frac{3 + \sqrt{5}}{4}$ | 2     | <p>2 marks for correct solution</p> <p>1 mark for finding the sum but not rationalising the denominator or equivalent merit</p> |



| Question | Working and answer                                                                                                                                                                                                                                                                                                                         | Marks    | Mark Allocation                                                                                                                                                                                                |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 18.      | $\int_0^{\frac{\pi}{2}} \sin\left(\frac{x}{2}\right) dx$ $= \left[ -2\cos\left(\frac{x}{2}\right) \right]_0^{\frac{\pi}{2}}$ $= \left( -2\cos\left(\frac{\pi}{2}\right) - (-2\cos 0) \right)$ $= -2\cos\left(\frac{\pi}{4}\right) + 2$ $= -2 \times \frac{1}{\sqrt{2}} + 2$ $= -\frac{2}{\sqrt{2}} + 2$ $= -\sqrt{2} + 2$ $= 2 - \sqrt{2}$ | <b>3</b> | <p>3 marks for correct solution</p> <p>2 marks for correct integration but error in substitution or equivalent merit</p> <p>1 mark for some form of integration with cos in the answer or equivalent merit</p> |
|          |                                                                                                                                                                                                                                                                                                                                            |          |                                                                                                                                                                                                                |

| Question |    | Working and answer                                                                                                                       | Marks    | Mark Allocation                                                                                                                                                   |
|----------|----|------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 19.      | a) | $x(t) = 6 - 2t + 8 \ln(t+3)$ <p>When <math>t=0</math></p> $x(0) = 6 - 0 + 8 \ln 3$ $= 6 + 8 \ln 3$                                       | <b>1</b> | 1 mark for correct solution                                                                                                                                       |
|          | b) | $x(t) = 6 - 2t + 8 \ln(t+3)$ $x'(t) = -2 + \frac{8}{t+3}$ $x''(t) = -\frac{8}{(t+3)^2}$                                                  | <b>2</b> | <p>2 marks for correct expressions for <math>x'(t)</math> and <math>x''(t)</math></p> <p>1 mark for progress toward <math>x'(t)</math> or <math>x''(t)</math></p> |
|          | c) | $x'(t) = -2 + \frac{8}{t+3}$ $0 = -2 + \frac{8}{t+3}$ $2 = \frac{8}{t+3}$ $2(t+3) = 8$ $2t+6 = 8$ $2t = 2$ $t = 1$ <p>After 1 second</p> | <b>1</b> | 1 mark for correct solution or equivalent merit                                                                                                                   |
|          |    |                                                                                                                                          |          |                                                                                                                                                                   |

| Question | Working and answer                                                                                                                                                                                                                             | Marks    | Mark Allocation                                                                                                                   |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------|
| 20. a)   | $P(t) = 100e^{kt}$ <p>When the time is 3 hours, <math>t = 180</math> mins, <math>P(180) = 3\,660</math></p> $3\,660 = 100e^{180k}$ $36.6 = e^{180k}$ $\ln(36.6) = 180k$ $k = \frac{\ln(36.6)}{180}$ $= 0.02000$ $k = 0.02 \text{ (1 sig fig)}$ | <b>2</b> | <p>2 marks for correct solution</p> <p>1 mark for finding <math>k</math> but using the value of 3 not 180 or equivalent merit</p> |
| 20. b)   | $P(t) = 100e^{0.02t}$ $P'(t) = 2e^{0.02t}$ <p>when <math>t = 180</math></p> $P'(180) = 2e^{0.02 \times 180}$ $= 73.1964$ <p>Rate of change = 73 bacteria/minute</p>                                                                            | <b>2</b> | 1 mark for correct solution or equivalent merit                                                                                   |

| Question | Working and answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks | Mark Allocation                                                                                                                                                                                 |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 20. c)   | $P'(t) = 2e^{0.02t}$ <p>Since <math>e^x &gt; 0</math> for all <math>x</math>, <math>P'(t) &gt; 0</math> for all <math>t</math></p> <p>So the rate of change is always positive and the population is always increasing</p> $P''(t) = 0.02 \times 2e^{0.02t}$ $= 0.04e^{0.02t} \text{ which is } > 0 \text{ for all } t.$ <p><math>\therefore</math> the rate of change of the population is always increasing.</p> <p>So the population is increasing at an increasing rate.</p> <p>Although the rate of increase is quite slow to begin with, (73 Bact/hr after 3 hrs is relatively slow) it will always increase unless some form of control is introduced.</p> | 2     | <p>2 marks for showing both derivatives are positive and <b>any</b> valid explanation relating to control that follows from this.</p> <p>1 mark for correct derivatives or equivalent merit</p> |

| Question |    | Working and answer                                                                                                                                                                       | Marks    | Mark Allocation           |
|----------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------|
| 21.      | a) | <p>Future <i>V</i> Interest Factor for 8% for 10 years = 14.487</p> <p>For an investment of \$5000 per year,</p> <p>Future Value = <math>14.487 \times 5000</math></p> <p>= \$72 435</p> | <b>1</b> | 1 mark for correct answer |
|          | b) | <p>Present <i>V</i> Interest Factor for 8% for 10 <i>years</i> = 6.710</p> <p>Present value = <math>6.710 \times 5000</math></p> <p>= \$33 550</p>                                       | <b>1</b> | 1 mark for correct answer |





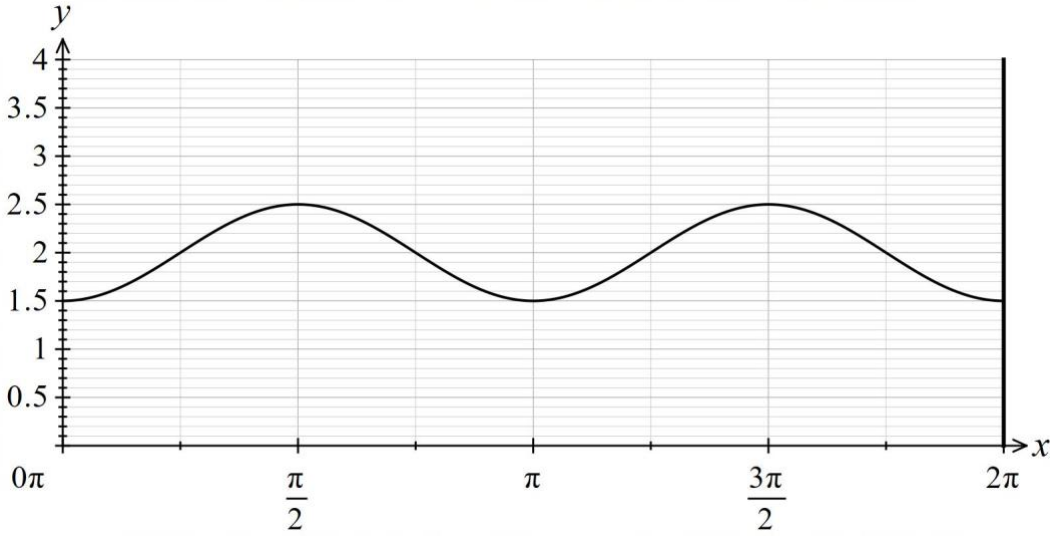
| Question | Working and answer                                                                                                                                                                                                                                                                                                                                                                   | Marks    | Mark Allocation                                                                                                                     |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------|
| 23.      | <p>a) As <math>f(x)</math> is a parabola with <math>a &lt; 0</math></p> <p>mode will occur at <math>x = -\frac{b}{2a}</math></p> $b = -\frac{3}{1\,372} \times -16 = \frac{48}{1\,372}$ $a = -\frac{3}{1\,372} \times 1 = -\frac{3}{1\,372}$ $x = \frac{-\frac{48}{1\,372}}{2 \times \frac{-3}{1\,372}}$ $x = 8$ <p><math>\therefore</math> the mode is <math>M = 8</math> hours</p> | <b>2</b> | <p>2 marks for correct solution</p> <p>1 mark for using the axis of symmetry formula with following error, or equivalent merit.</p> |

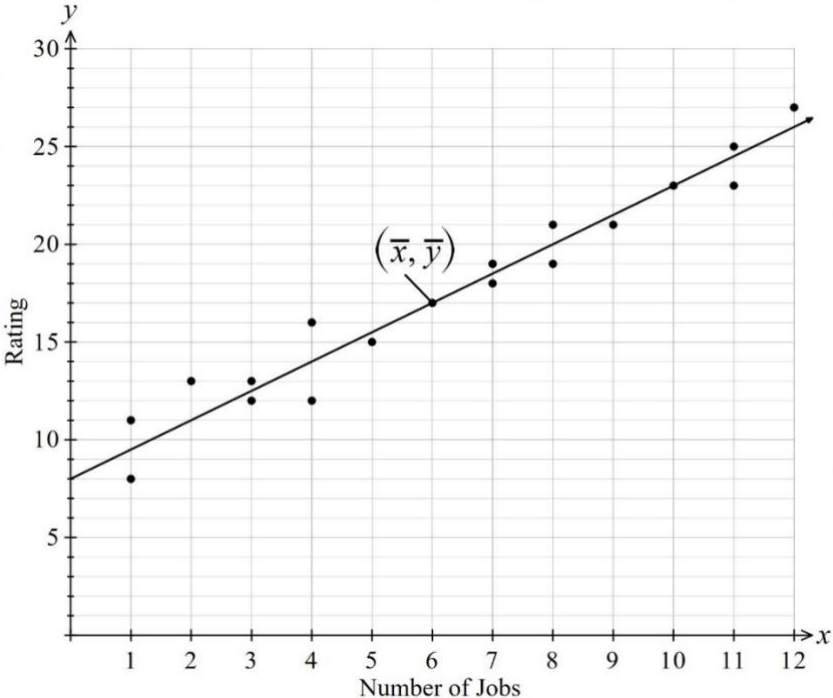


| Question |    | Working and answer                                                                                                                                                                                                                                                                                                                                                                                           | Marks | Mark Allocation                                                                                                                                   |
|----------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 23       | b) | $P(5 \leq X \leq 8) = P(X \leq 8) - P(X \leq 5)$ <p>So</p> $\int_5^8 -\frac{3}{1\,372}(x^2 - 16x + 15) \, dx$ $= \left[ -\frac{3}{1\,372} \left( \frac{x^3}{3} - 8x^2 + 15x \right) \right]_5^8$ $= -\frac{3}{1\,372} \left( \left( -\frac{664}{3} \right) - \left( -\frac{250}{3} \right) \right)$ $= -\frac{3}{1\,372} \left( -\frac{414}{3} \right)$ $= 0.30174927$ $= 30\% \text{ (to nearest percent)}$ | 2     | <p>2 marks for correct solution</p> <p>1 for make some progress finding an integral between 5 and 8 even if not correctly or equivalent merit</p> |
|          |    |                                                                                                                                                                                                                                                                                                                                                                                                              |       |                                                                                                                                                   |

| Question |    | Working and answer                                                                                                                                                                                                                                                         | Marks    | Mark Allocation                                                                                 |
|----------|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------|
| 24.      | a) | $y = k \sin[a(x + b)] + c$<br>$h(x) = 0.5 \sin\left(2x - \frac{\pi}{2}\right) + 2$<br>$= 0.5 \sin\left[2\left(x - \frac{\pi}{4}\right)\right] + 2$<br>Amplitude $k = 0.5$<br>Period $= \frac{2\pi}{2} = \pi$<br>Phase Shift $= \frac{\pi}{4}$ to the right<br>Centre $= 2$ | <b>2</b> | 2 marks for four correct values or equivalent merit<br><br>1 mark for at least 2 correct values |

| Question |    | Working and answer                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks | Mark Allocation                                                                                                             |
|----------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------|
| 24.      | b) | $2 = 0.5 \sin\left(2x - \frac{\pi}{2}\right) + 2$ $0 = 0.5 \sin\left(2x - \frac{\pi}{2}\right)$ $0 = \sin\left(2x - \frac{\pi}{2}\right)$ $2x - \frac{\pi}{2} = 0, \pi, 2\pi, 3\pi, 4\pi$ $2x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \frac{9\pi}{2}$ $x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}, \frac{9\pi}{4}$ $x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4} \text{ for } [0, 2\pi]$ | 2     | <p>2 marks for correct solution</p> <p>1 mark for progress towards finding values of <math>x</math> or equivalent merit</p> |

| Question | Working and answer                                                                                                        | Marks | Mark Allocation                                                                                                                                                                                |
|----------|---------------------------------------------------------------------------------------------------------------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 24       | <p data-bbox="219 268 264 304">(c)</p>  | 2     | <p data-bbox="1559 268 2056 347">2 marks for correct graph from values found in part (b)</p> <p data-bbox="1559 448 2018 528">1 mark for minor error in one of the properties of the graph</p> |

| Question | Working and answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks    | Mark Allocation                                                                                                                                                                                                                                                                                                                                                                        |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 25.      | <p>a) The gradient of the line is <math>m = 1.5</math>.<br/> <math>(\bar{x}, \bar{y}) = (6, 17)</math> lies on the line.<br/> Equation is of the form <math>y - y_1 = m(x - x_1)</math><br/> <math>y - 17 = 1.5(x - 6)</math><br/> <math>y - 17 = 1.5x - 9</math><br/> <math>y = 1.5x + 8</math><br/> So gradient is 1.5 and y intercept is 8 and we also know that it passes through <math>(\bar{x}, \bar{y}) = (6, 17)</math>.<br/> Line is sketched on the graph below.</p>  | <b>3</b> | <p>3 marks for correct equation and correct line shown on the graph.</p> <p>2 marks for correct equation only or equivalent merit</p> <p>1 mark for correct graph from an incorrect equation or equivalent merit</p> <p>Students who use Statistics Mode may get a slightly different answer, but rounding should give the same result (or very close depending on their rounding)</p> |

| Question |    | Working and answer                                                                                                                                                                                                                                                                                                       | Marks    | Mark Allocation                                                         |
|----------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------|
| 25.      | b) | <p>If <math>x = 16</math>, <math>y = 1.5 \times 16 + 8</math><br/> <math>= 32</math></p>                                                                                                                                                                                                                                 | <b>1</b> | 1 mark for correct answer                                               |
| 25.      | c) | <p>Since the rating is out of 40, it would not be possible to predict a rating greater than 40<br/> <math>40 = 1.5 \times x + 8</math><br/> <math>32 = 1.5 \times x</math><br/> <math>x = \frac{32}{1.5} = 21.33333333</math><br/>           So for 22 jobs the prediction would be greater than the maximum rating.</p> | <b>1</b> | 1 mark for a correct explanation, not required to give an actual value. |
|          |    |                                                                                                                                                                                                                                                                                                                          |          |                                                                         |

| Question | Working and answer                                                                                                                                                                                                                                                                    | Marks    | Mark Allocation                                                                                                                        |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------|
| 26.      | $\begin{aligned} \text{Difference} &= \left( 2 + \frac{t^2}{t+1} \right) - \left( 1 + \frac{1}{t+1} \right) \\ &= \frac{2(t+1) + t^2}{(t+1)} - \frac{(t+1) + 1}{t+1} \\ &= \frac{2t + 2 + t^2 - t - 2}{t+1} \\ &= \frac{t^2 + t}{t+1} \\ &= \frac{t(t+1)}{t+1} \\ &= t \end{aligned}$ | <b>2</b> | <p>2 marks for the correct solution</p> <p>1 mark for trying to find the difference between the two equations or equivalent merit.</p> |
|          |                                                                                                                                                                                                                                                                                       |          |                                                                                                                                        |

| Question | Working and answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks    | Mark Allocation                                                                                                                                                                                                                                                                                                                                                               |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 27.      | $z = \frac{x - \mu}{\frac{\sigma}{\sqrt{n}}}$ $\mu = 125$ $\sigma = 15$ <p>When <math>x = 143</math></p> $z_1 = \frac{143 - 125}{\frac{15}{\sqrt{25}}}$ $= 1.2$ <p>Probability from table = 0.8849</p> <p>When <math>x = 137</math></p> $z_2 = \frac{137 - 125}{\frac{15}{\sqrt{25}}}$ $= 0.8$ <p>Probability from table = 0.7881</p> <p>Probability between 143 and 137 = <math>0.8849 - 0.7881</math></p> $= 0.0968$ <p>Number between 143 and 137 = <math>0.0968 \times 1250</math></p> $= 121 \text{ students}$ | <b>5</b> | <p>5 marks for correct answer</p> <p>4 marks for the correct final probability or equivalent merit</p> <p>3 marks for final probability found using one incorrect value from table or equivalent merit</p> <p>2 marks for 2 correct z-scores and attempt to use them with probability table or equivalent merit</p> <p>1 mark for one correct z-score or equivalent merit</p> |
|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |                                                                                                                                                                                                                                                                                                                                                                               |
| 28.      | <p>a) Perimeter = radius of sector + length of semi-circle + length of arc of sector</p> $P = r + r\theta + \left( \frac{1}{2} \times \pi \left( \frac{r}{2} \right)^2 \right)$ $= r + r\theta + \frac{\pi r^2}{2}$                                                                                                                                                                                                                                                                                                 | <b>2</b> | <p>2 marks for the correct solution</p> <p>1 mark for progress towards finding the correct perimeter</p>                                                                                                                                                                                                                                                                      |





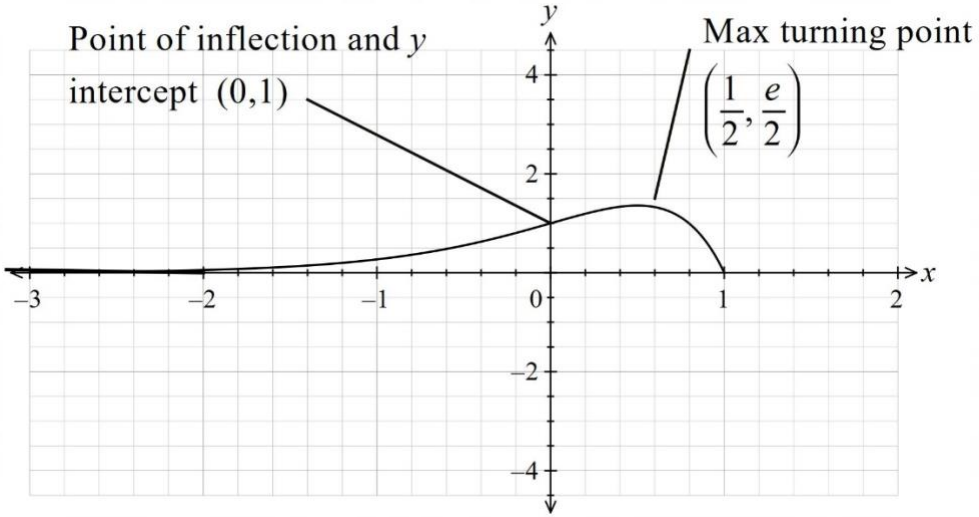
| Question |              | Working and answer                                                                                                                                                                                                                                           | Marks | Mark Allocation                                                                                                                                                                                     |
|----------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 28.      | b)<br>(cont) | $= r + \frac{2}{r} - \frac{\pi r}{4} + \frac{\pi r}{2}$ $= \frac{2}{r} + r - \frac{\pi r}{4} + \frac{2\pi r}{4}$ $= \frac{2}{r} + r + \frac{\pi r}{4}$ $= \frac{2}{r} + r \left( 1 + \frac{\pi}{4} \right)$                                                  |       | See Above                                                                                                                                                                                           |
| 28.      | c)           | $P = \frac{2}{r} + r \left( 1 + \frac{\pi}{4} \right)$ $= 2r^{-1} + r \left( 1 + \frac{\pi}{4} \right)$ $\frac{dP}{dr} = -2r^{-2} + \left( 1 + \frac{\pi}{4} \right)$ $= -\frac{2}{r^2} + 1 + \frac{\pi}{4}$ $\frac{d^2P}{dr^2} = 4r^{-3}$ $= \frac{4}{r^3}$ | 3     | <p>3 marks for correctly showing required result and finding <math>\theta</math></p> <p>2 marks for correctly showing required result oem</p> <p>1 mark for progress toward required result oem</p> |

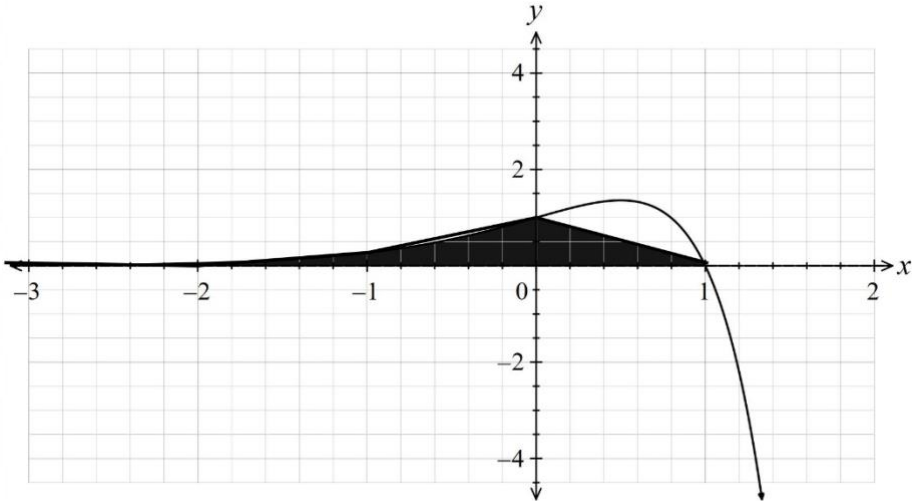
| Question         | Working and answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks | Mark Allocation |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------|
| 28. c)<br>(cont) | <p>Stationary points occur when</p> $\frac{dP}{dr} = 0$ $0 = -\frac{2}{r^2} + 1 + \frac{\pi}{4}$ $\frac{2}{r^2} = 1 + \frac{\pi}{4}$ $\frac{4 + \pi}{4} = \frac{2}{r^2}$ $r^2(4 + \pi) = 8$ $r^2 = \frac{8}{4 + \pi}$ $r = \sqrt{\frac{8}{4 + \pi}}$ <p>When <math>r = \sqrt{\frac{8}{4 + \pi}} \quad \frac{d^2P}{dr^2} &gt; 0</math></p> <p><math>\therefore</math> a minimum turning point when</p> $r^2 = \frac{8}{4 + \pi}$ <p><math>\therefore</math> the least perimeter will occur at</p> $r^2 = \frac{8}{4 + \pi}$ |       | See Above       |

|     |              |                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                        |  |           |
|-----|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------|
| 28. | c)<br>(cont) | $\theta = \frac{8 - \pi r^2}{4r^2}$ <p>when <math>r^2 = \frac{8}{4 + \pi}</math></p> $\theta = \frac{8 - \pi \left( \frac{8}{4 + \pi} \right)}{4 \left( \frac{8}{4 + \pi} \right)}$ $= \frac{8 - \frac{8\pi}{4 + \pi}}{\frac{32}{4 + \pi}}$ $= \frac{8(4 + \pi) - 8\pi}{\frac{32}{4 + \pi}}$ $= \frac{32}{4 + \pi}$ $= 1 \text{ radian}$ $\theta = \frac{180}{\pi}$ $= 57.29577951$ $= 57^\circ$ | $\theta = \frac{8 - \pi r^2}{4r^2}$ $= \frac{8}{4r^2} - \frac{\pi r^2}{4r^2}$ $= \frac{2}{r^2} - \frac{\pi}{4}$ <p>When <math>r^2 = \frac{8}{4 + \pi}</math></p> $\theta = 2 \frac{4 + \pi}{8} - \frac{\pi}{4}$ $= \frac{8 + 2\pi}{8} - \frac{2\pi}{8}$ $= \frac{8}{8}$ $= 1^c$ $= \frac{180}{\pi}^\circ$ $= 57^\circ$ |  | See Above |
|-----|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------|

| Question | Working and answer                                                                                                                                                                                                                                                                                                                                                                                          | Marks    | Mark Allocation                                                                                                                                                                                                                              |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 29.      | <p>a)</p> $f(x) = e^{2x}(1-x)$ $u = e^{2x} \quad v = 1-x$ $u' = 2e^{2x} \quad v' = -1$ $f'(x) = 2e^{2x}(1-x) - e^{2x}$ $= 2e^{2x} - 2xe^{2x} - e^{2x}$ $= e^{2x} - 2xe^{2x}$ $= e^{2x}(1-2x) \quad ***$<br>$f'(x) = e^{2x}(1-2x)$ $u = e^{2x} \quad v = 1-2x$ $u' = 2e^{2x} \quad v' = -2$ $f''(x) = 2e^{2x}(1-2x) - 2e^{2x}$ $= 2e^{2x} - 4xe^{2x} - 2e^{2x}$ $= -4xe^{2x} \quad ***$ $f''(x) = -4xe^{2x}$ | <b>4</b> | <p>4 marks for finding turning point and nature and finding inflexion and testing</p><br><p>3 marks for finding turning point and nature oem</p><br><p>2 marks for finding turning point oem</p><br><p>1 mark for correct derivative oem</p> |

|     |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |           |
|-----|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------|
| 29. | a)<br>(cont) | <p>stat pts <math>f'(x) = 0</math></p> $0 = e^{2x}(1 - 2x)$ $\therefore e^{2x} = 0 \quad \text{or} \quad 1 - 2x = 0$ $e^{2x} \neq 0 \quad x = \frac{1}{2}$ <p>when <math>x = \frac{1}{2}</math> <math>f(x) = e^1 \left(1 - \frac{1}{2}\right) = \frac{e}{2}</math></p> <p>classify in <math>f''(x)</math></p> $f''(x) = -4xe^{2x}$ $f''\left(\frac{1}{2}\right) = -2e < 0$ <p><math>\therefore</math> concave down</p> <p><math>\therefore</math> max turning pt <math>\left(\frac{1}{2}, \frac{e}{2}\right)</math></p> <p>Inflections</p> $f''(x) = 0$ $0 = -4xe^{2x}$ $\therefore -4x = 0 \quad \text{or} \quad e^{2x} = 0$ $x = 0 \quad e^{2x} \neq 0$ <p>when <math>x = 0</math> <math>f(x) = e^0(1 - 0) = 1</math></p> <p><math>\therefore</math> possible inflection (0, 1)</p> <p>check concavity change</p> $f''(-1) = -4 \times -1e^{-2} > 0$ $f''(1) = -4 \times 1e^2 < 0$ <p><math>\therefore</math> concavity change</p> <p><math>\therefore</math> inflection at (0, 1)</p> |  | See above |
|-----|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------|

| Question | Working and answer                                                                 | Marks    | Mark Allocation                                                                                                                                                                                                                           |
|----------|------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 29. b)   |  | <b>2</b> | <p>2 marks for correct graph with inflection and turning point shown</p> <p>1 mark for using working from part <i>a</i> with small errors or not showing both points</p> <p>0 marks if only plotting coordinates from the given table</p> |
| 29. c)   | Curve is concave down between $[0, 1]$                                             | <b>1</b> | 1 mark for correct solution or equivalent merit                                                                                                                                                                                           |

| Question | Working and answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks    | Mark Allocation                                                                                         |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------|
| 29. d)   | <p>The approximation will be less than the actual area, as the trapezium formed between <math>x = 0</math> and <math>x = 1</math> (actually a triangle) will be considerably smaller than the actual area, due to the downward concavity of the curve, and while the area of the trapezia from <math>x = -3</math> to <math>x = 0</math> will be larger than the actual area, the difference is only very small as shown on shading below.</p>  | <b>1</b> | 1 mark for any explanation which mentions trapezia being below curve and area less than the exact area. |



| Question |    | Working and answer                                                                                                                                                                                                                                                                                                                                                                 | Marks | Mark Allocation                                                                                                                                                       |
|----------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 30.      | a) | $\cos\theta = \frac{AD}{AB} \therefore AD = AB\cos\theta$ $\tan\theta = \frac{BC}{AB} \therefore BC = AB\tan\theta$ $\cos\theta = \frac{AB}{AC} \therefore \sec\theta = \frac{AC}{AB}, \therefore AC = AB\sec\theta$ $6AD + BC = 5AC$ $6(AB\cos\theta) + AB\tan\theta = 5(AB\sec\theta)$ $AB(6\cos\theta + \tan\theta) = AB(5\sec\theta)$ $6\cos\theta + \tan\theta = 5\sec\theta$ | 2     | <p>2 marks for the correct solution</p> <p>1 mark for finding trig ratios to express lengths of sides AD, BC and AC even if not fully correct or equivalent merit</p> |
|          | b) | $6\cos\theta + \tan\theta = 5\sec\theta$ $6\cos\theta + \frac{\sin\theta}{\cos\theta} = \frac{5}{\cos\theta}$ $6\cos^2\theta + \sin\theta = 5$ $6(1 - \sin^2\theta) + \sin\theta = 5$ $6 - 6\sin^2\theta + \sin\theta = 5$ $6\sin^2\theta - \sin\theta - 1 = 0$                                                                                                                    | 2     | <p>2 marks for correct solution</p> <p>1 mark for making some progress and using trig ratios to simplify the equation or equivalent merit</p>                         |

| Question |    | Working and answer                                                                                                                                                                                                                                                                                                                                                                                        | Marks    | Mark Allocation                                                                                                |
|----------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------|
| 30.      | c) | <p>Let <math>x = \sin\theta</math></p> $6\sin^2\theta - \sin\theta - 1 = 0$ $6x^2 - x - 1 = 0$ $6x^2 - 3x + 2x - 1 = 0$ $3x(2x - 1) + 1(2x - 1) = 0$ $(2x - 1)(3x + 1) = 0$ $\therefore x = \frac{1}{2}, -\frac{1}{3}$ <p>So</p> $\sin\theta = \frac{1}{2} \text{ or } \sin\theta = -\frac{1}{3}$ <p><math>0 \leq \theta \leq 90^\circ</math> as right-angled triangle</p> $\therefore \theta = 30^\circ$ | <b>2</b> | <p>2 marks for correct solution</p><br><br><br><br><p>1 mark for solving the quadratic or equivalent merit</p> |

| Question | Working and answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Marks | Mark Allocation                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31.      | $P = 500\,000, r = 5 \div 100 \div 12 = \frac{1}{240}$ $A_1 = 500\,000 \left(1 + \frac{1}{240}\right)^1 - 4\,000$ $A_2 = A_1 \left(1 + \frac{1}{240}\right)^1 - 4\,000$ $= 500\,000 \left(1 + \frac{1}{240}\right)^2 - 4\,000 \left(1 + \frac{1}{240}\right) - 4\,000$ $A_3 = A_2 \left(1 + \frac{1}{240}\right)^1 - 4\,000$ $= 500\,000 \left(1 + \frac{1}{240}\right)^3 - 4\,000 \left(1 + \frac{1}{240}\right)^2 - 4\,000 \left(1 + \frac{1}{240}\right) - 4\,000$ $A_n = 500\,000 \left(1 + \frac{1}{240}\right)^n - 4\,000 \left[ \left(1 + \frac{1}{240}\right)^{n-1} + \left(1 + \frac{1}{240}\right)^{n-2} + \dots + 1 \right]$ | 5     | <p>5 marks for the correct solution</p> <p>4 marks for significant progress towards the solution but small errors in process or calculations or equivalent merit</p> <p>3 marks for getting to an equation for <math>A_n</math> even if not entirely correct or equivalent merit</p> <p>2 marks for progress towards an equation for <math>A_n</math> or equivalent merit</p> <p>1 mark for setting up the equation for <math>A_1</math> or equivalent merit</p> |

| Question | Working and answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks | Mark Allocation |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------|
| 31.      | <p data-bbox="208 256 293 320">(cont)</p> <p data-bbox="293 256 1386 304">But <math>A_n = 0</math> when the money is gone.</p> $0 = 500\,000 \left( \frac{241}{240} \right)^n - 4\,000 \left[ \left( 1 + \frac{241}{240} + \dots + \left( \frac{241}{240} \right)^{n-1} \right) \right]$ <p data-bbox="293 504 1386 616"><math>\left[ \left( 1 + \frac{241}{240} + \dots + \left( \frac{241}{240} \right)^{n-1} \right) \right]</math> is a geometric series</p> $a = 1, r = \frac{241}{240}$ $S_n = \frac{a(r^n - 1)}{r - 1}$ $= \frac{1 \left( \left( \frac{241}{240} \right)^n - 1 \right)}{\frac{241}{240} - 1}$ $= \frac{\left( \frac{241}{240} \right)^n - 1}{\frac{1}{240}}$ |       | See Above       |

| Question | Working and answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks | Mark Allocation |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------|
| 31.      | <div>(cont)</div> $\therefore$ $500\,000\left(\frac{241}{240}\right)^n = 4\,000\frac{\left(\frac{241}{240}\right)^n - 1}{\frac{1}{240}}$ $\frac{6\,250}{3}\left(\frac{241}{240}\right)^n = 4\,000\left(\frac{241}{240}\right)^n - 4\,000$ $\frac{5\,750}{3}\left(\frac{241}{240}\right)^n = 4\,000$ $\ln\left(\frac{241}{240}\right)^n = \ln\left(\frac{48}{23}\right)$ $n = \frac{\ln\left(\frac{48}{23}\right)}{\ln\left(\frac{241}{240}\right)}$ $n = 176.93722293 \text{ months}$ <p>So Maisey's money will run out <math>25 + 14.75 = 39.75\text{years}</math><br/>ie 3 months before her 40th birthday</p> |       | See Above       |
|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |                 |